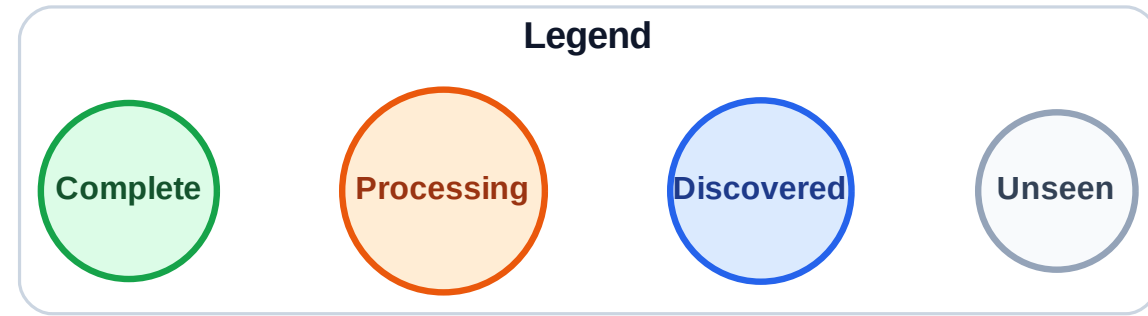


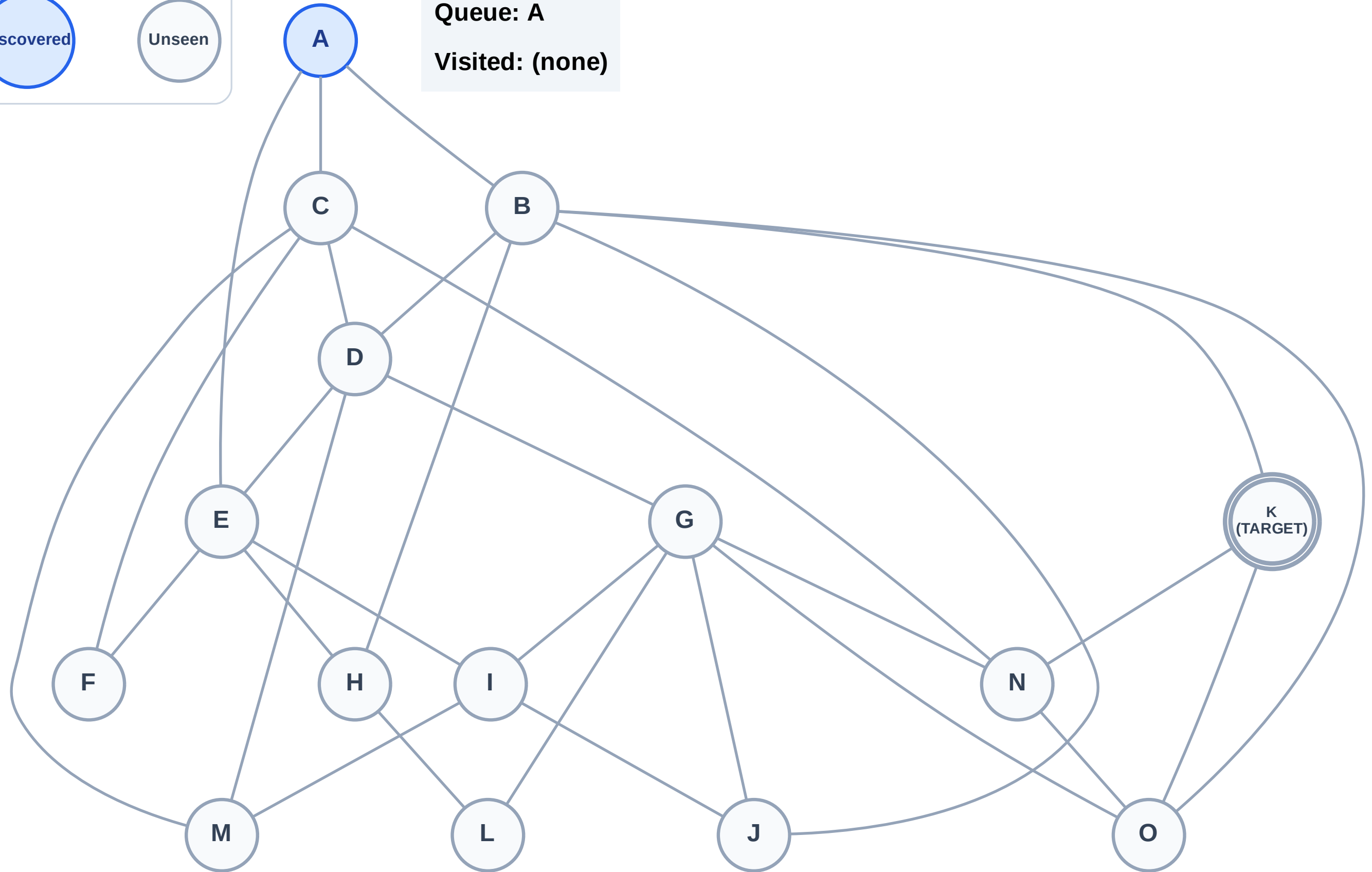
BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

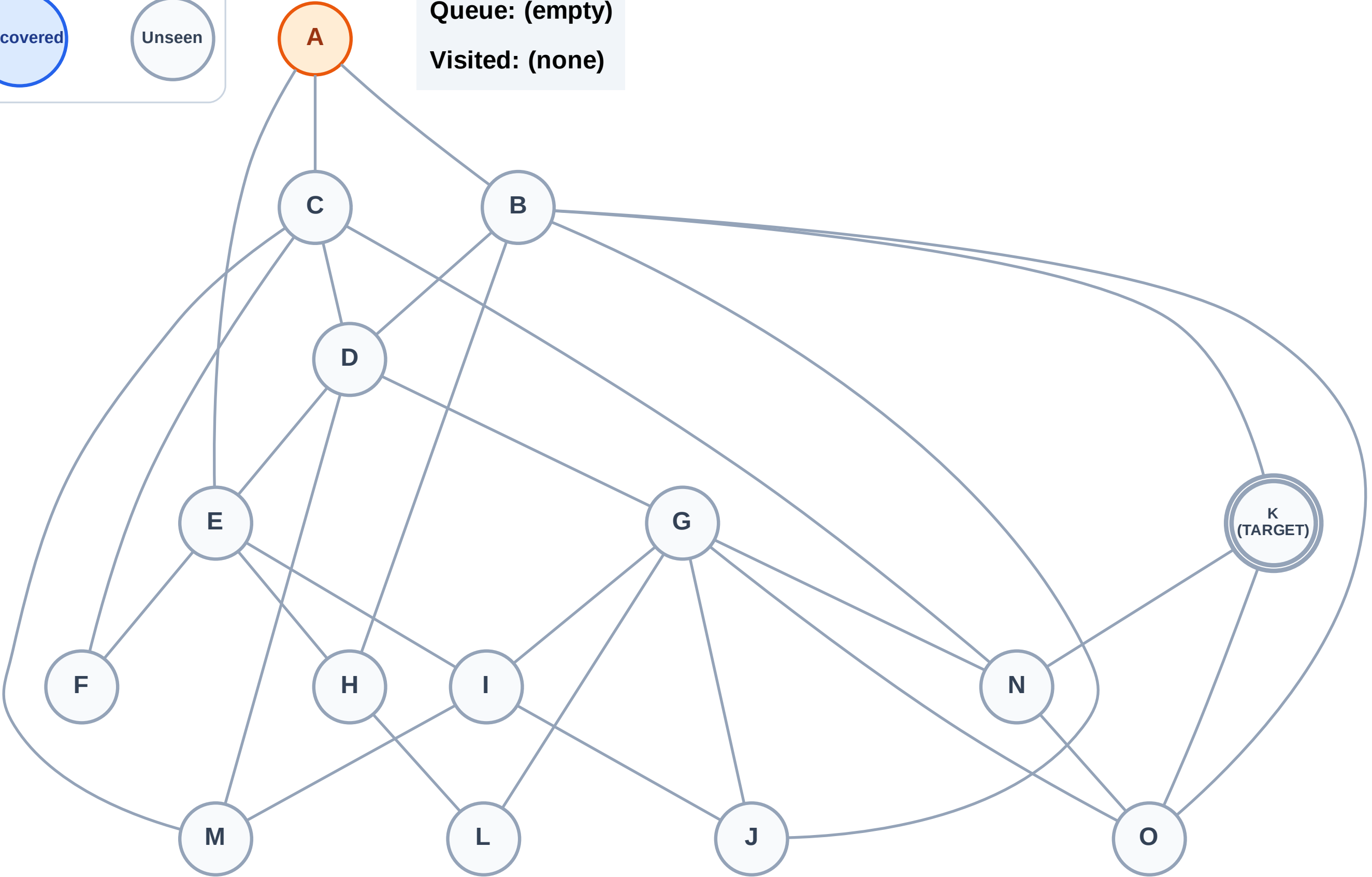
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, E from A

Legend

Complete

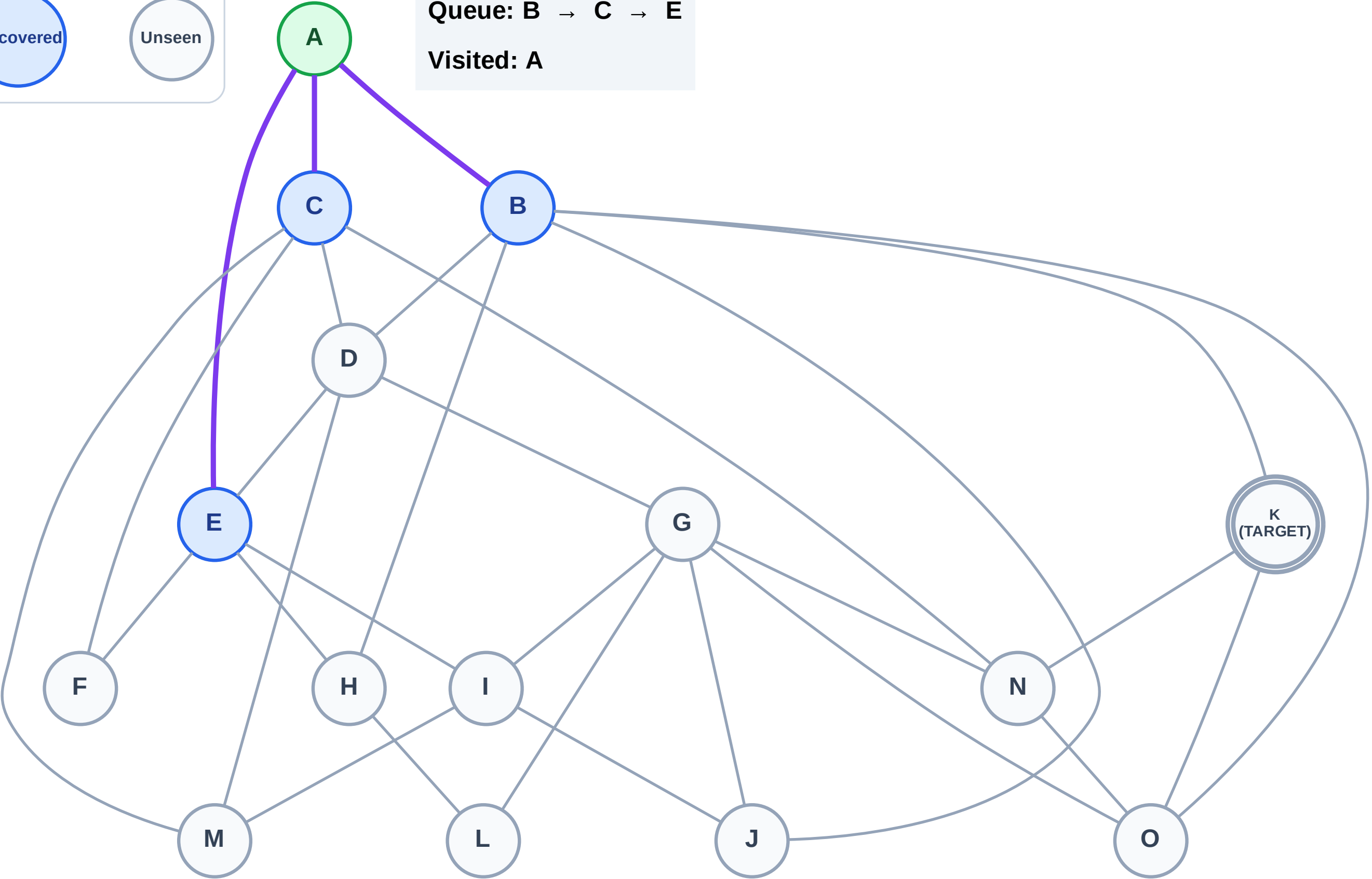
Processing

Discovered

Unseen

Queue: B → C → E

Visited: A



BFS Traversal

Step 4: Process node B

Legend

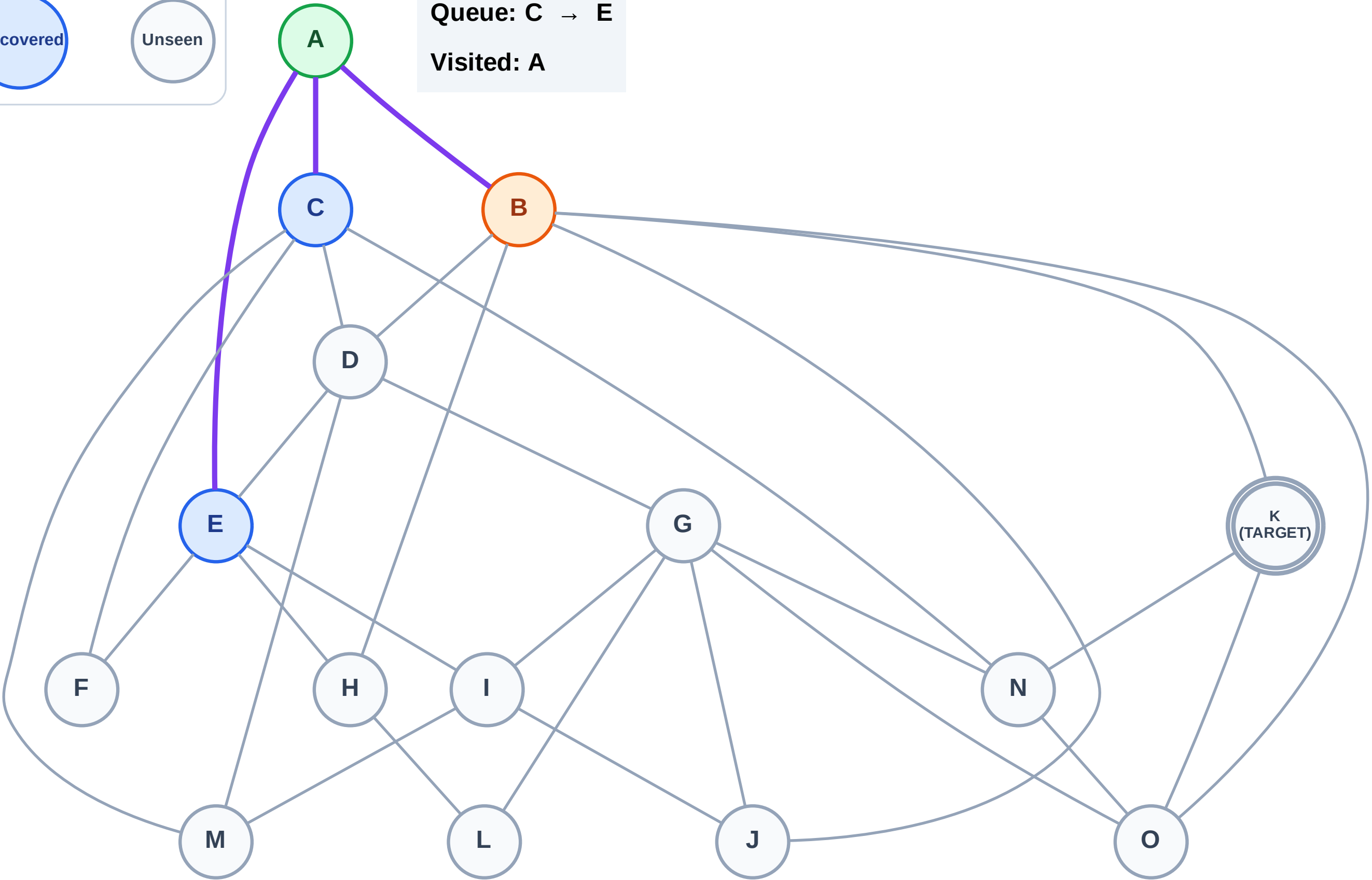
Complete

Processing

Discovered

Unseen

Queue: C → E
Visited: A



BFS Traversal

Step 5: Discover D, H, J, K, O from B

Legend

Complete

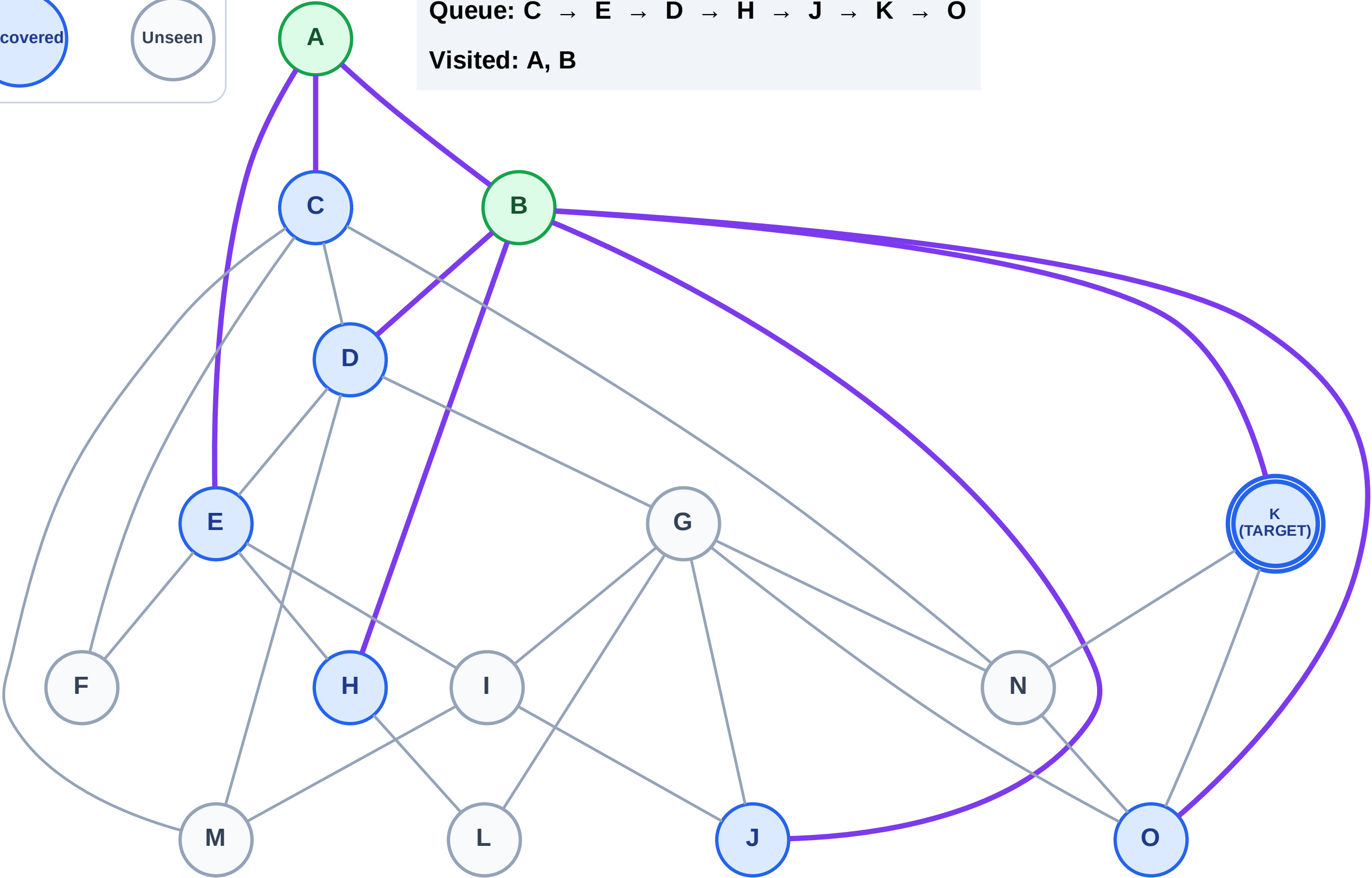
Processing

Discovered

Unseen

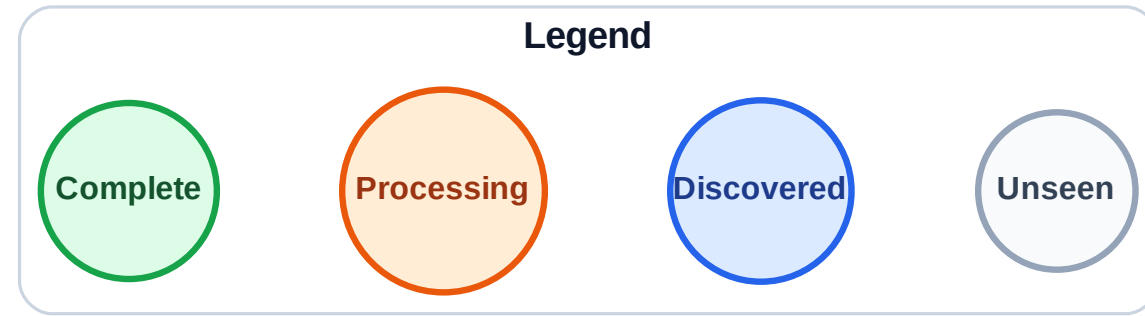
Queue: C → E → D → H → J → K → O

Visited: A, B

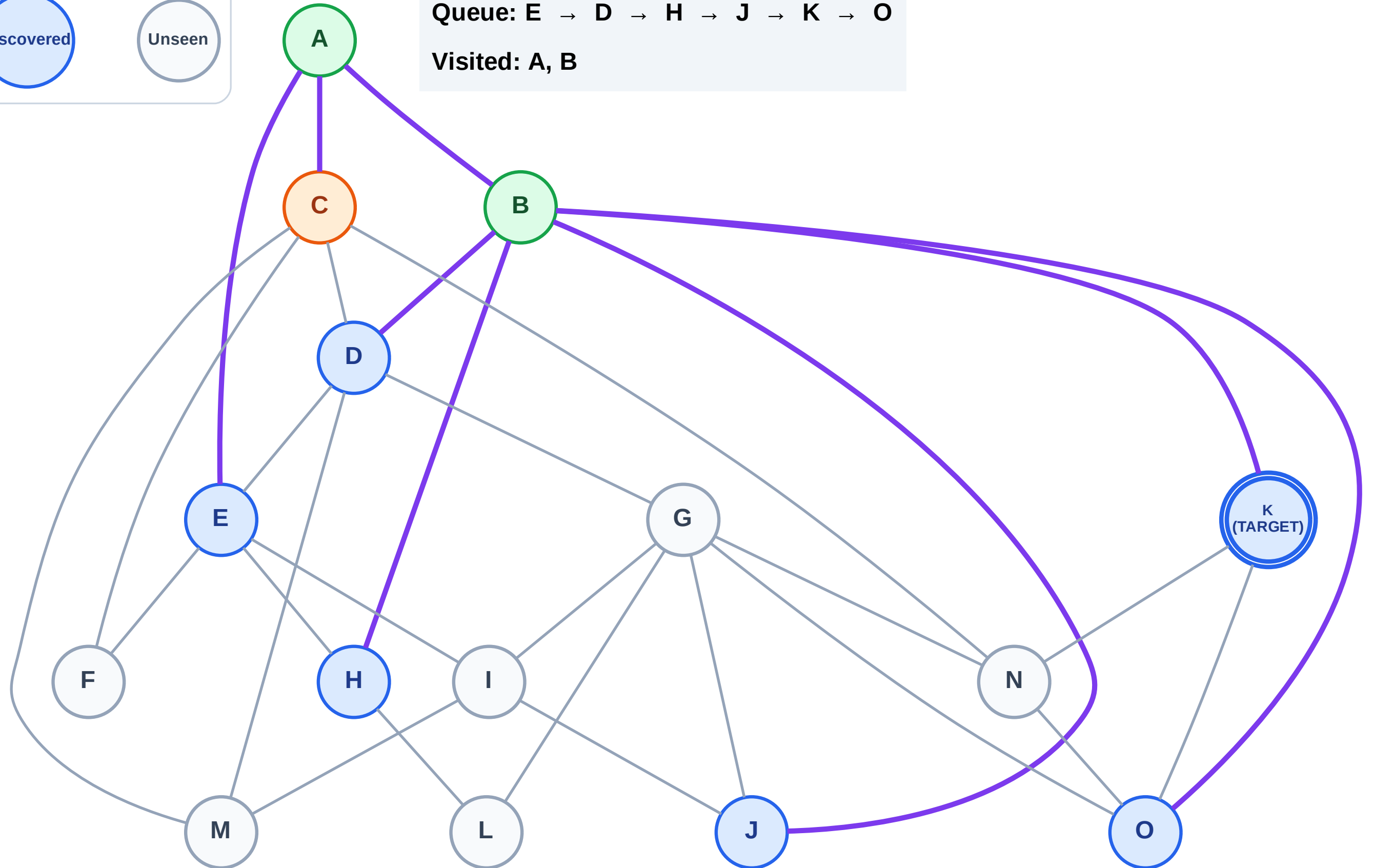


BFS Traversal

Step 6: Process node C



Queue: E → D → H → J → K → O
Visited: A, B



BFS Traversal

Step 7: Discover F, M, N from C

Legend

Complete

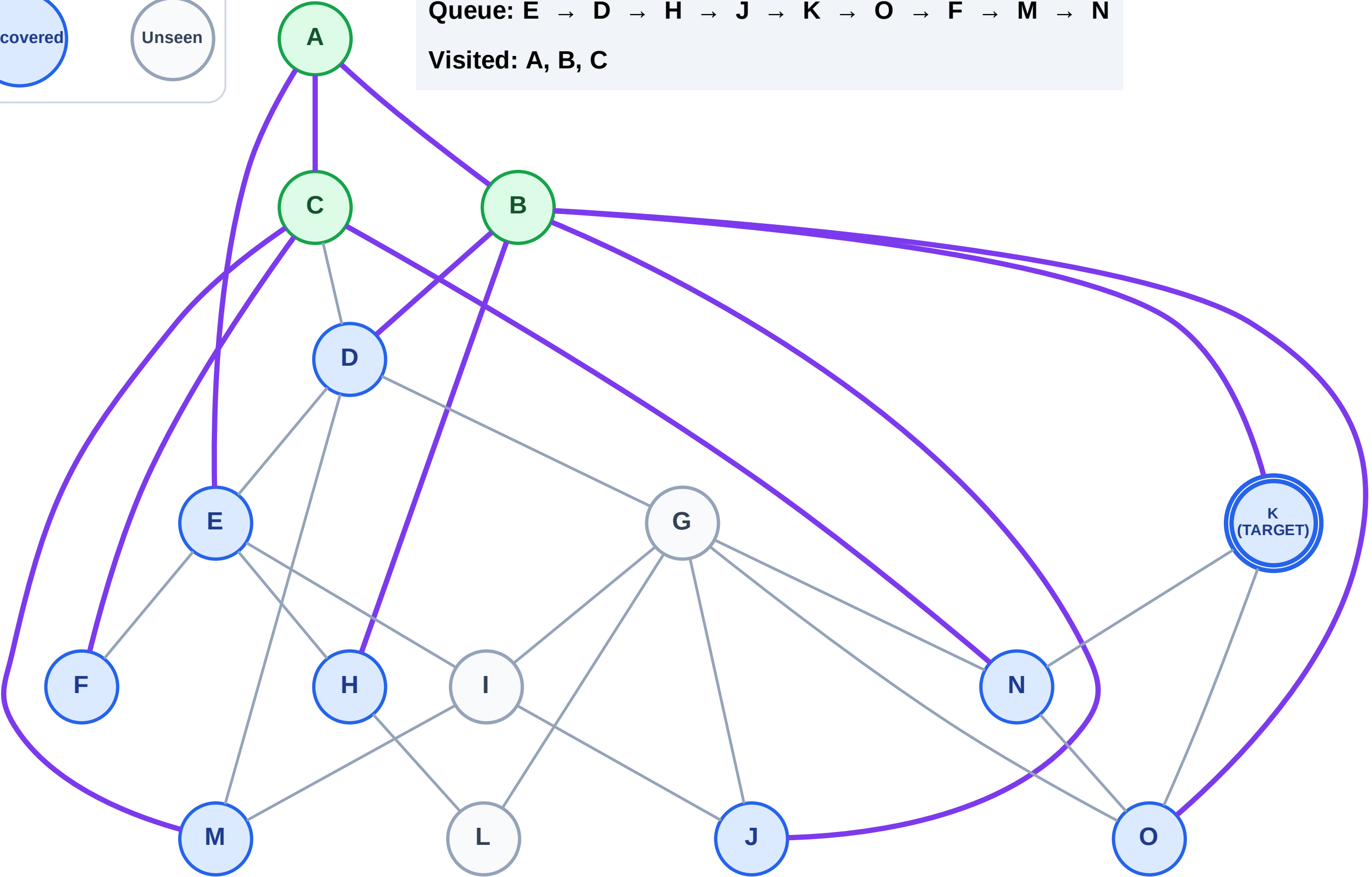
Processing

Discovered

Unseen

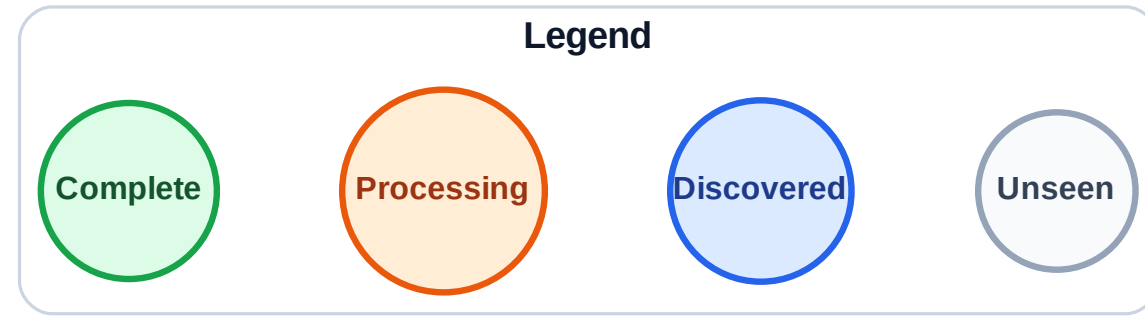
Queue: E → D → H → J → K → O → F → M → N

Visited: A, B, C



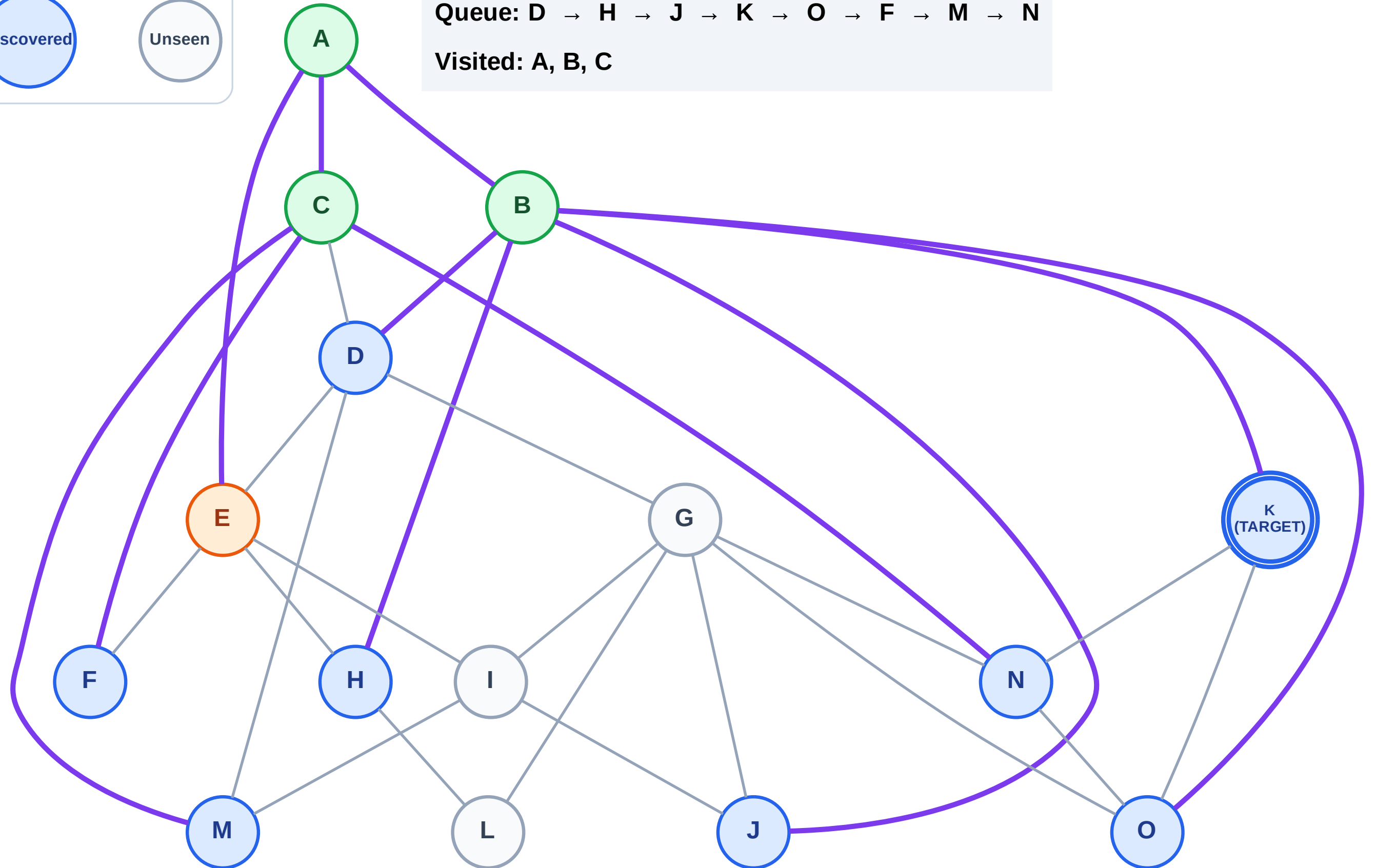
BFS Traversal

Step 8: Process node E



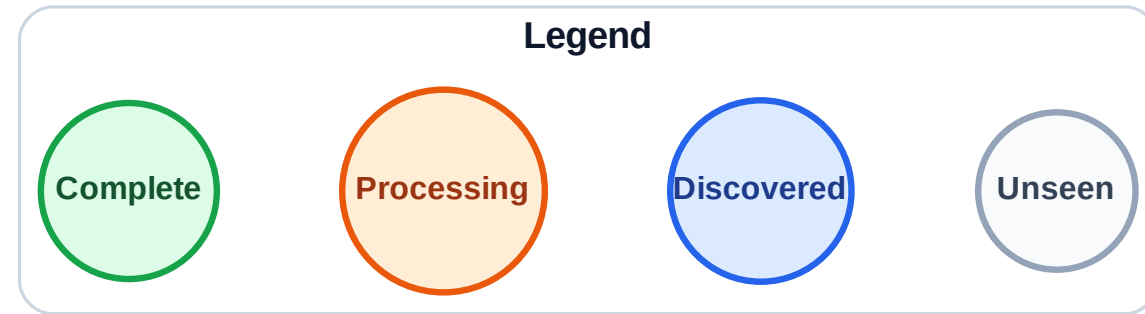
Queue: D → H → J → K → O → F → M → N

Visited: A, B, C



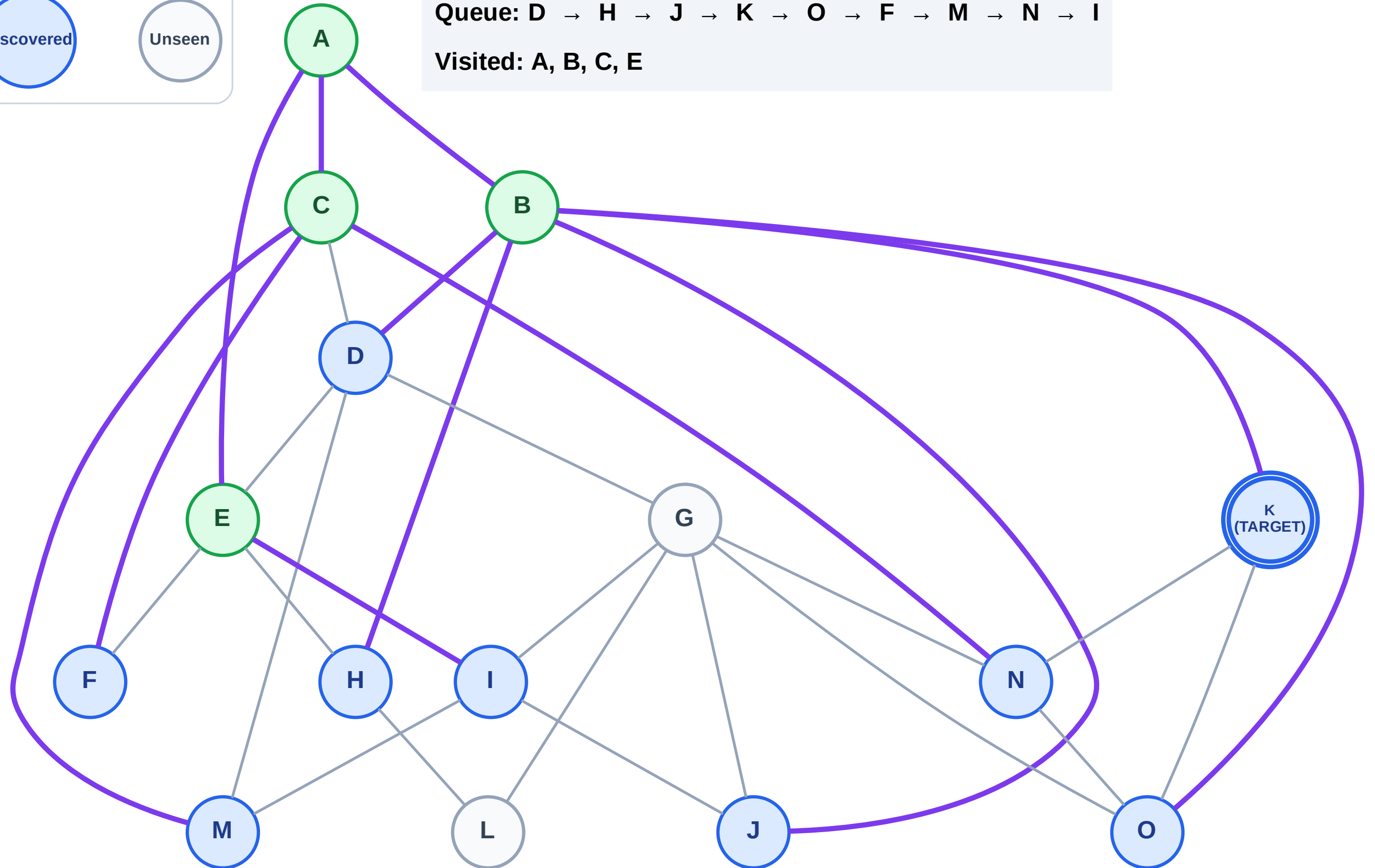
BFS Traversal

Step 9: Discover I from E



Queue: D → H → J → K → O → F → M → N → I

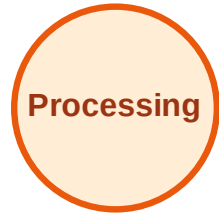
Visited: A, B, C, E



BFS Traversal

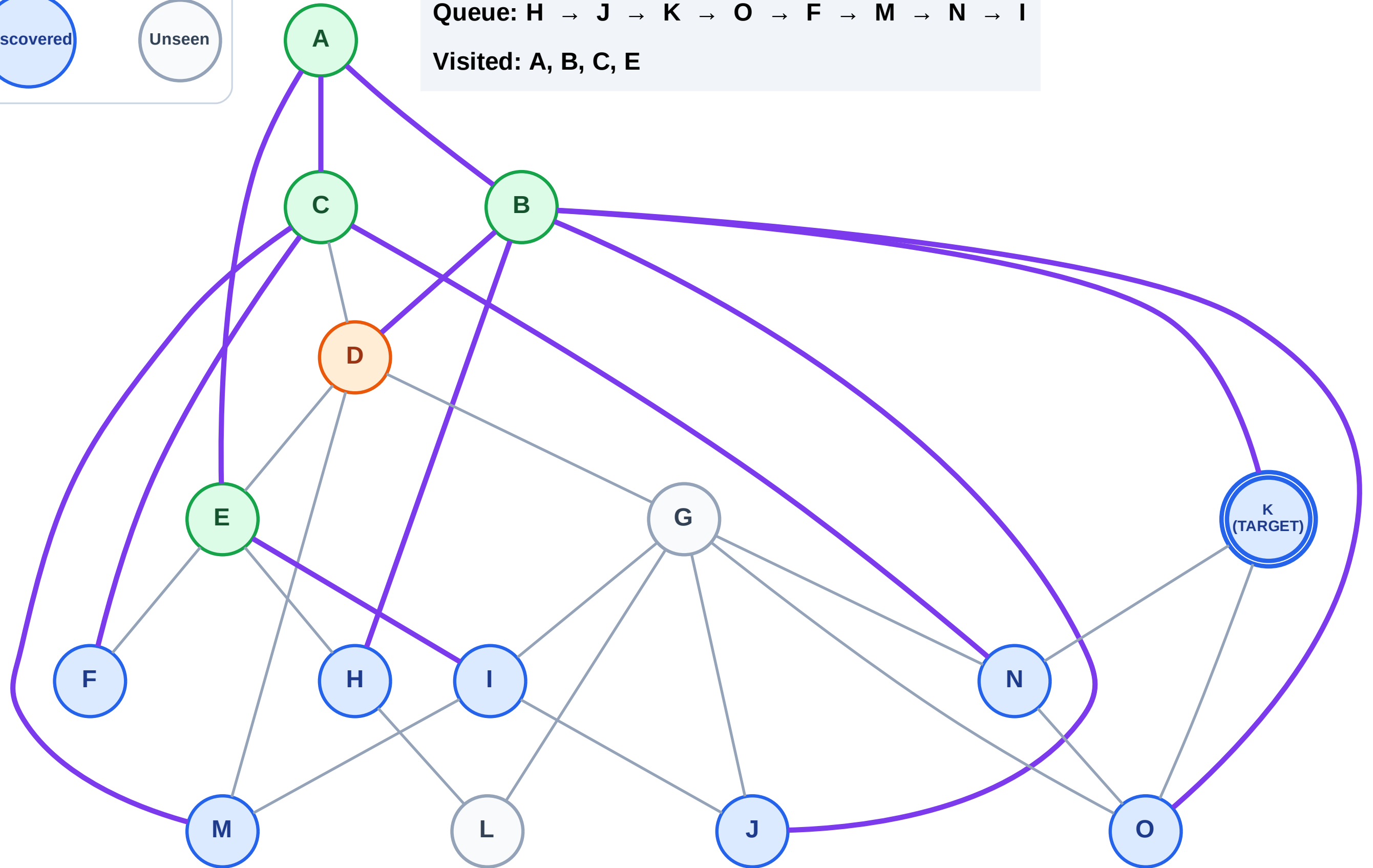
Step 10: Process node D

Legend



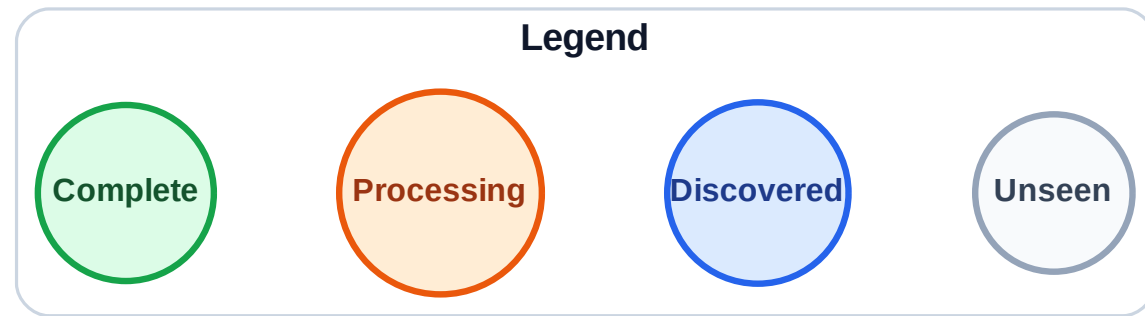
Queue: H → J → K → O → F → M → N → I

Visited: A, B, C, E



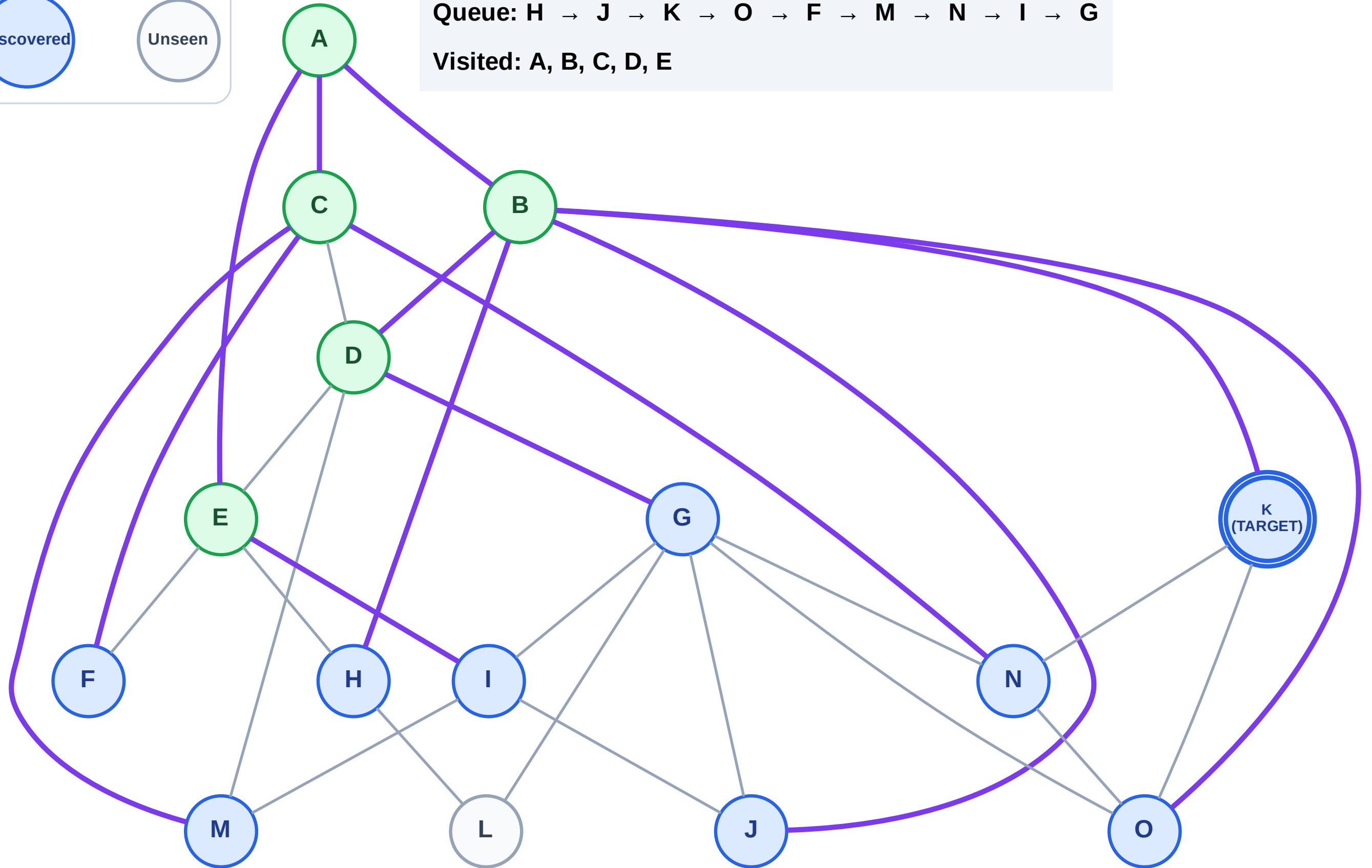
Step 11: Discover G from D

Step 11: Discover G from D



Queue: H → J → K → O → F → M → N → I → G

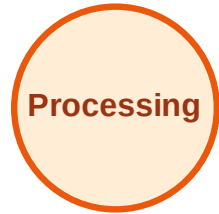
Visited: A, B, C, D, E



BFS Traversal

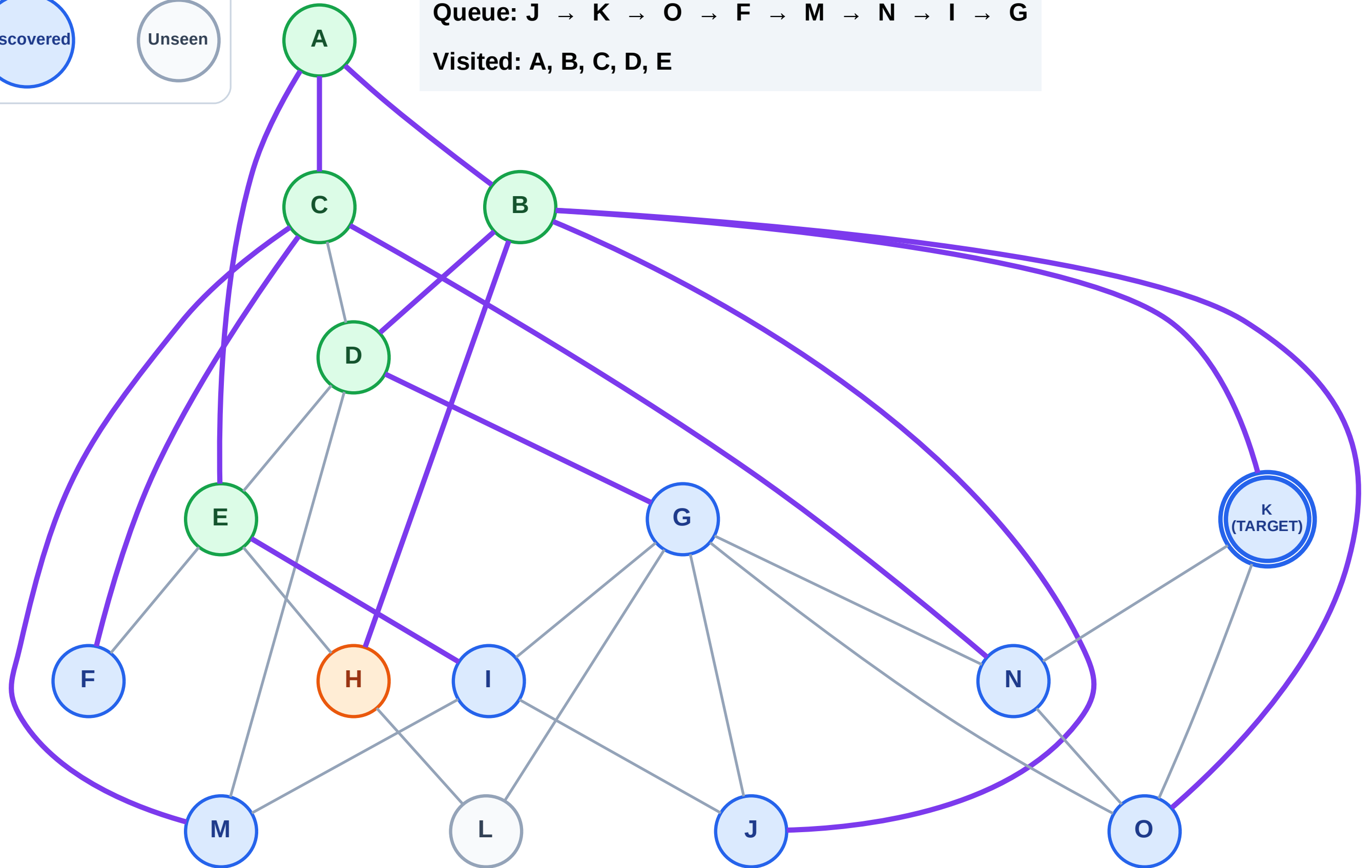
Step 12: Process node H

Legend



Queue: J → K → O → F → M → N → I → G

Visited: A, B, C, D, E



BFS Traversal

Step 13: Discover L from H

Legend

Complete

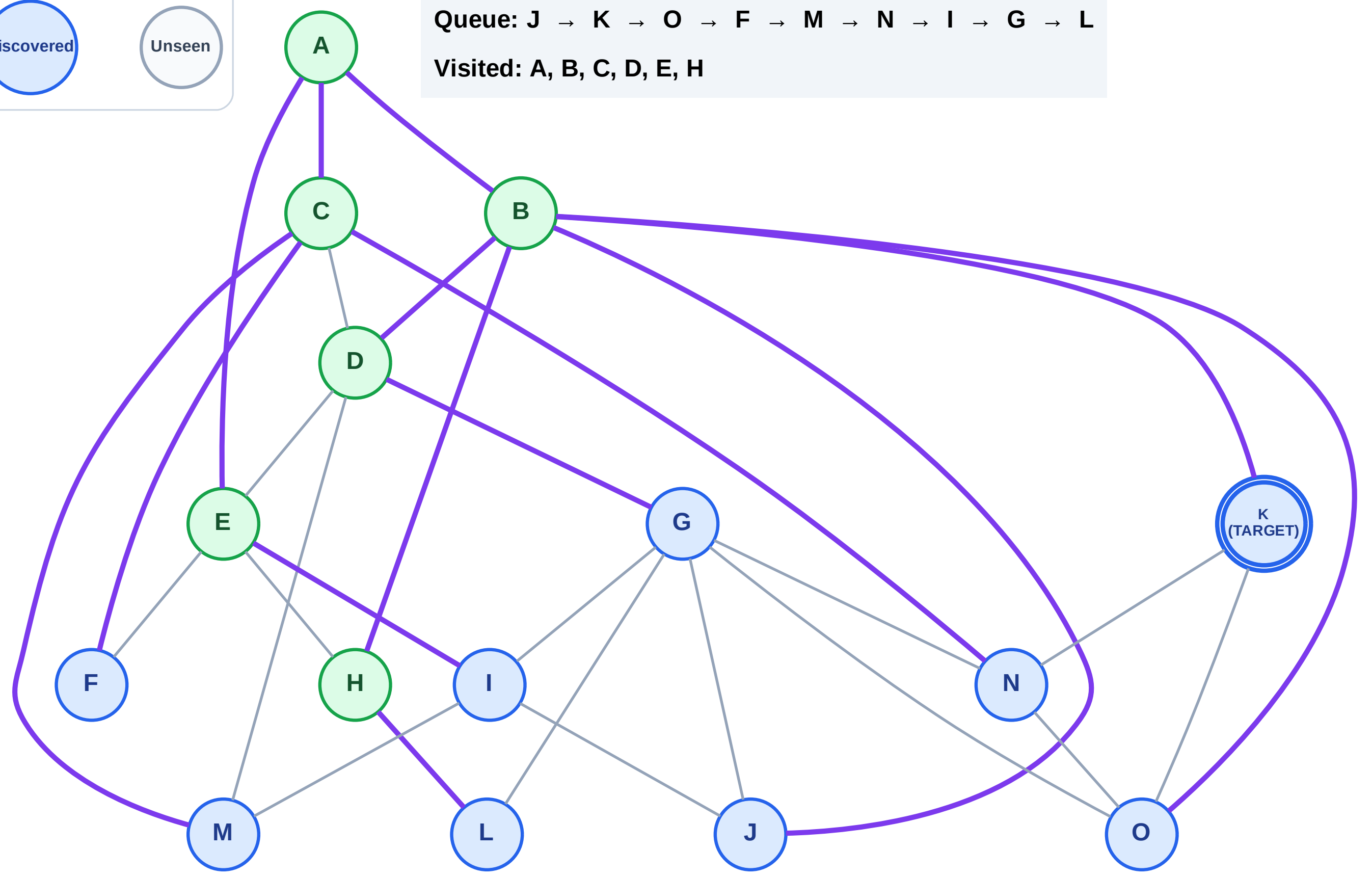
Processing

Discovered

Unseen

Queue: J → K → O → F → M → N → I → G → L

Visited: A, B, C, D, E, H



BFS Traversal

Step 14: Process node J

Legend

Complete

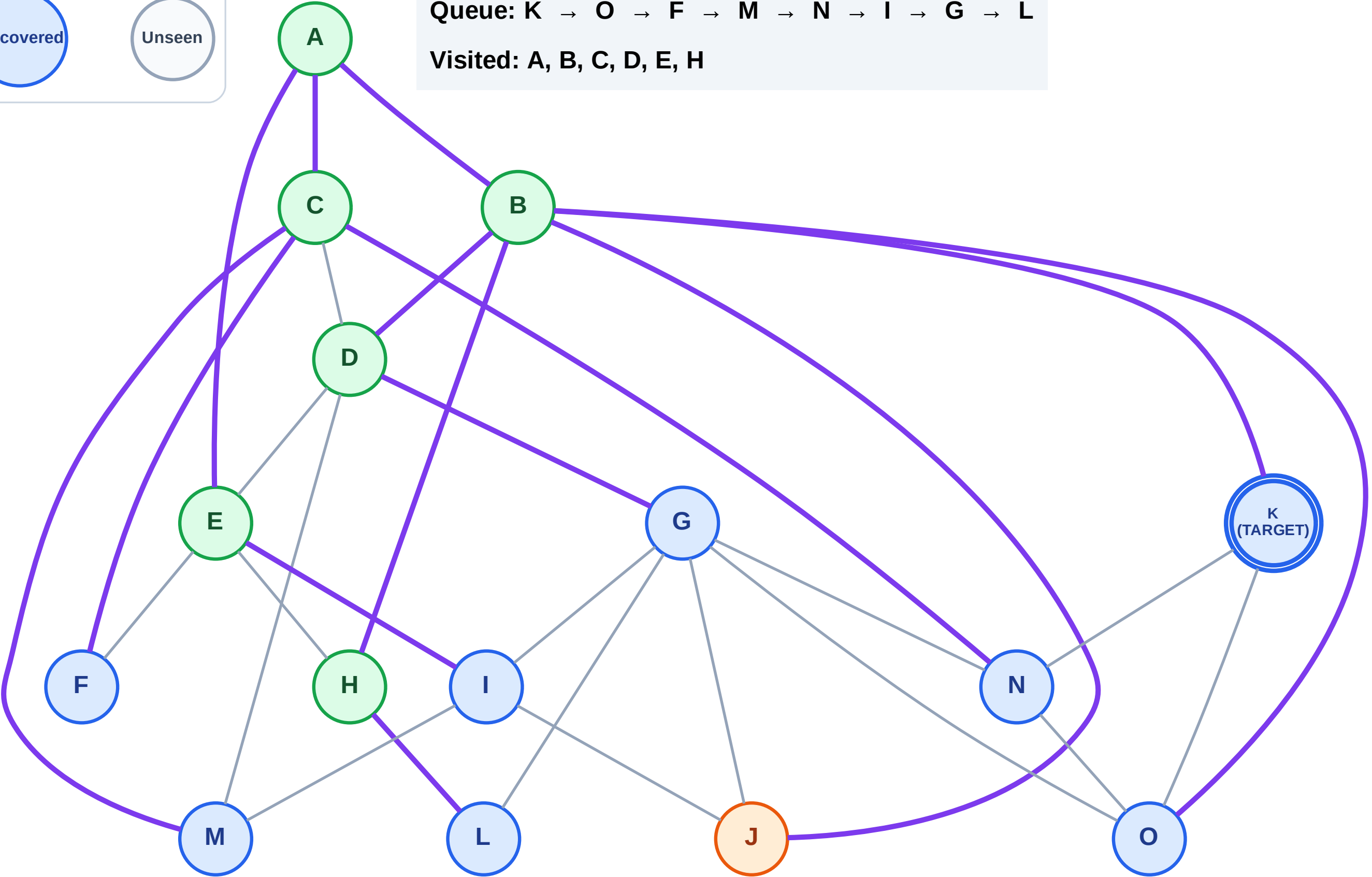
Processing

Discovered

Unseen

Queue: K → O → F → M → N → I → G → L

Visited: A, B, C, D, E, H



BFS Traversal

Step 15: No new neighbors from J

Legend

Complete

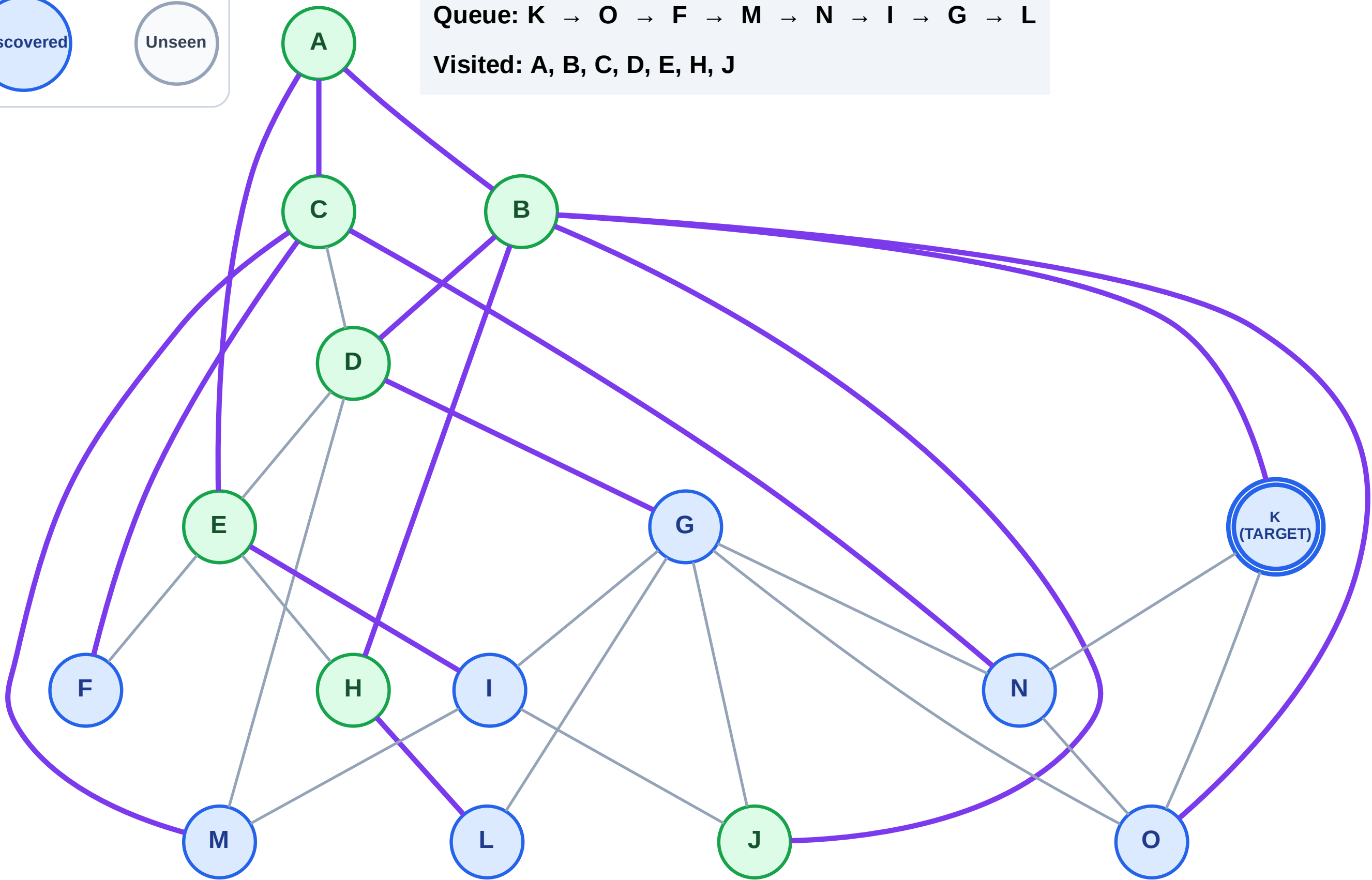
Processing

Discovered

Unseen

Queue: K → O → F → M → N → I → G → L

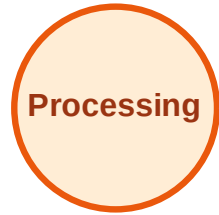
Visited: A, B, C, D, E, H, J



BFS Traversal

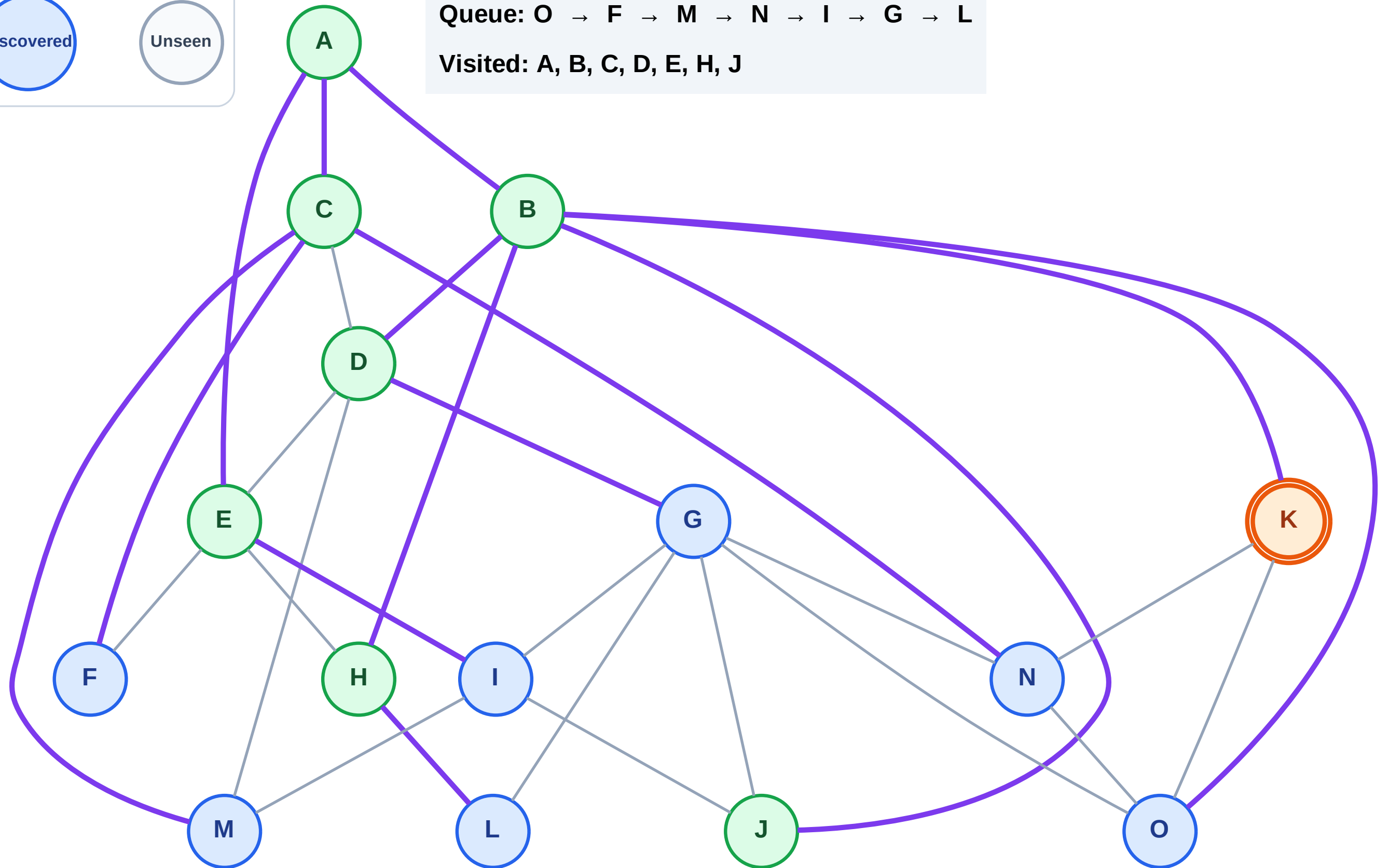
Step 16: Process node K

Legend



Queue: O → F → M → N → I → G → L

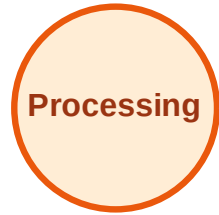
Visited: A, B, C, D, E, H, J



BFS Traversal

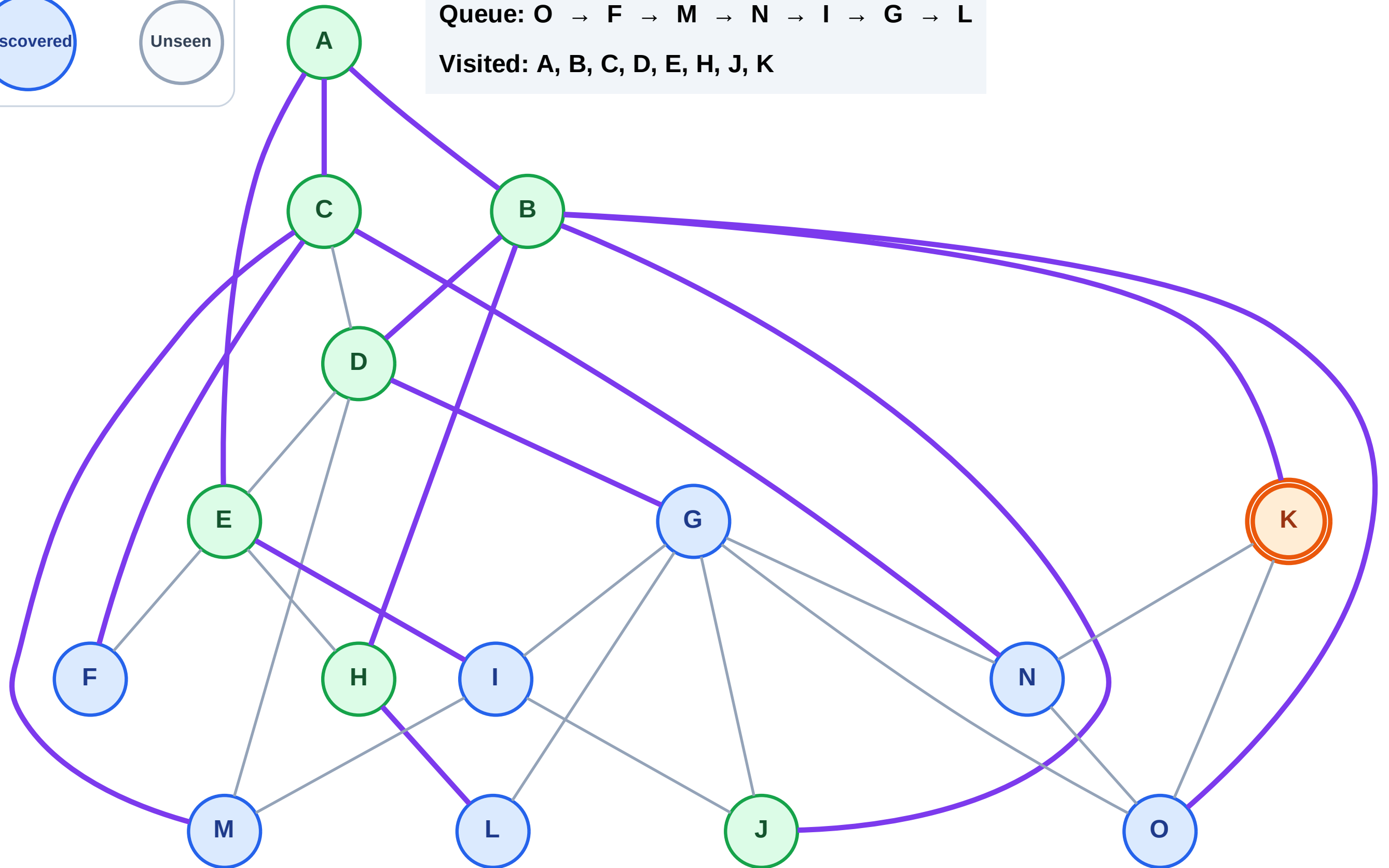
Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



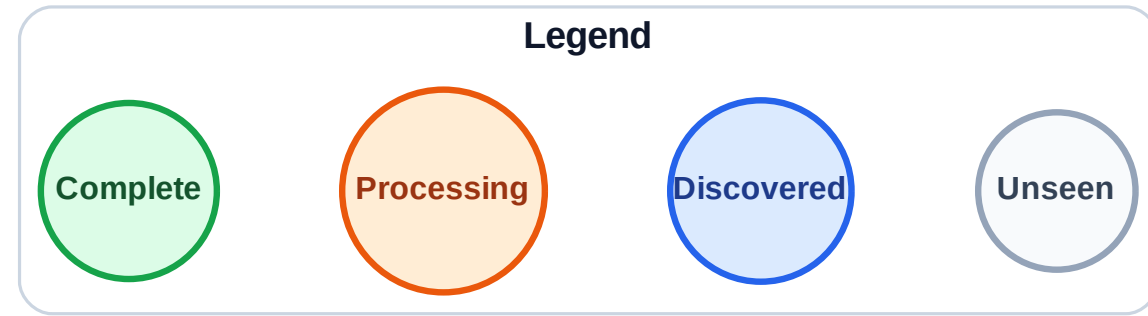
Queue: O → F → M → N → I → G → L

Visited: A, B, C, D, E, H, J, K

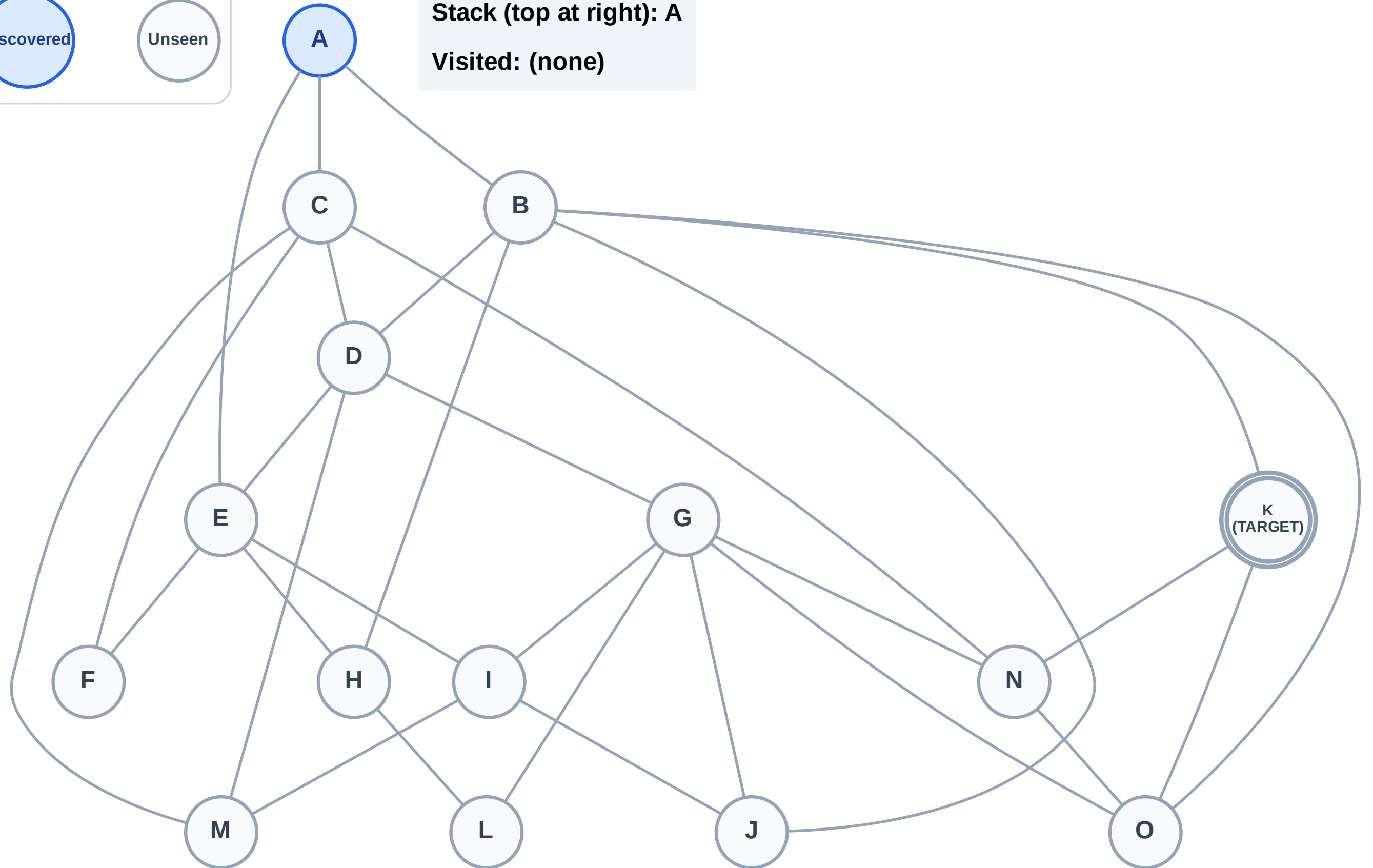


DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

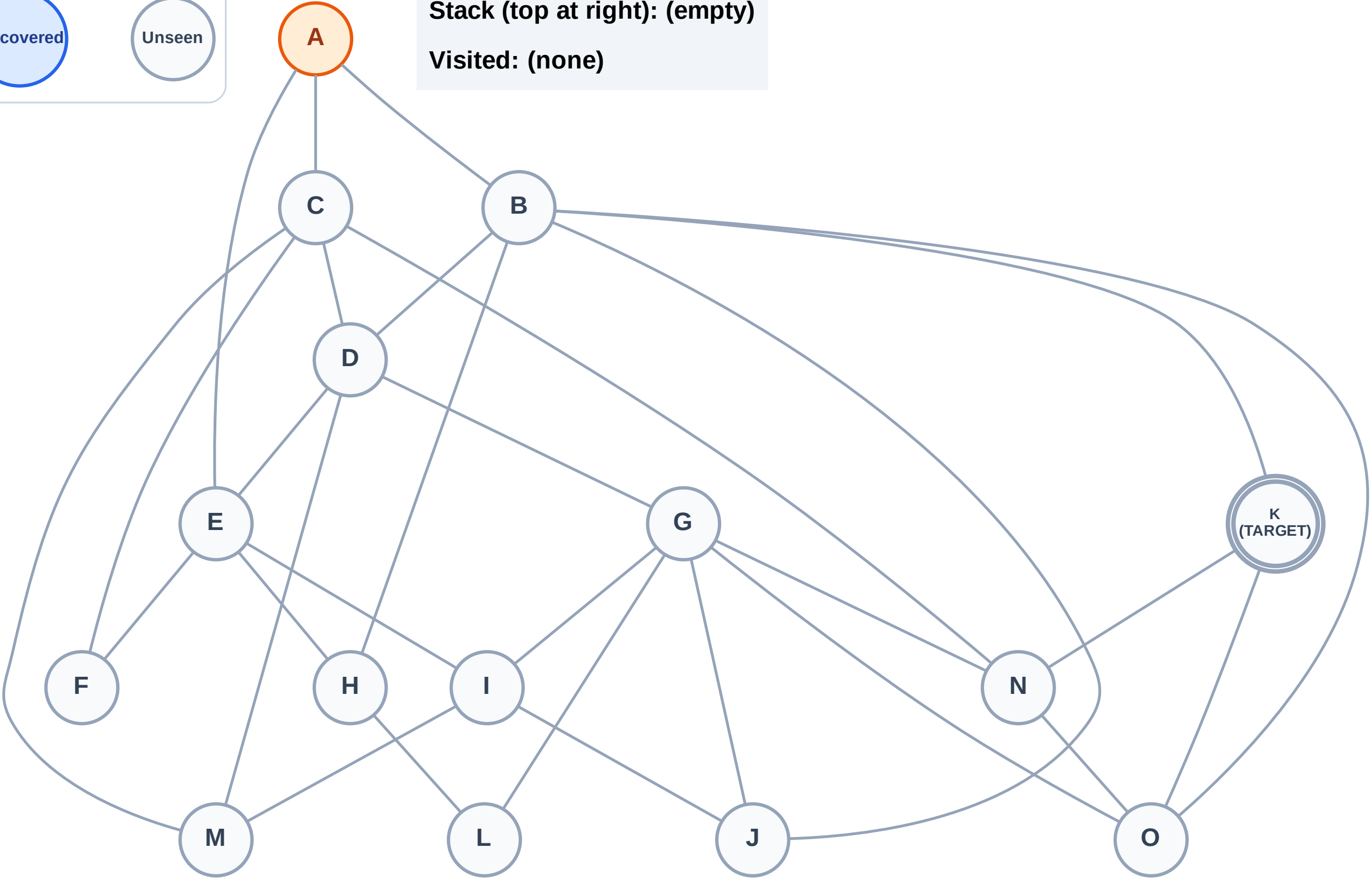
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover E, C, B from A

Legend

Complete

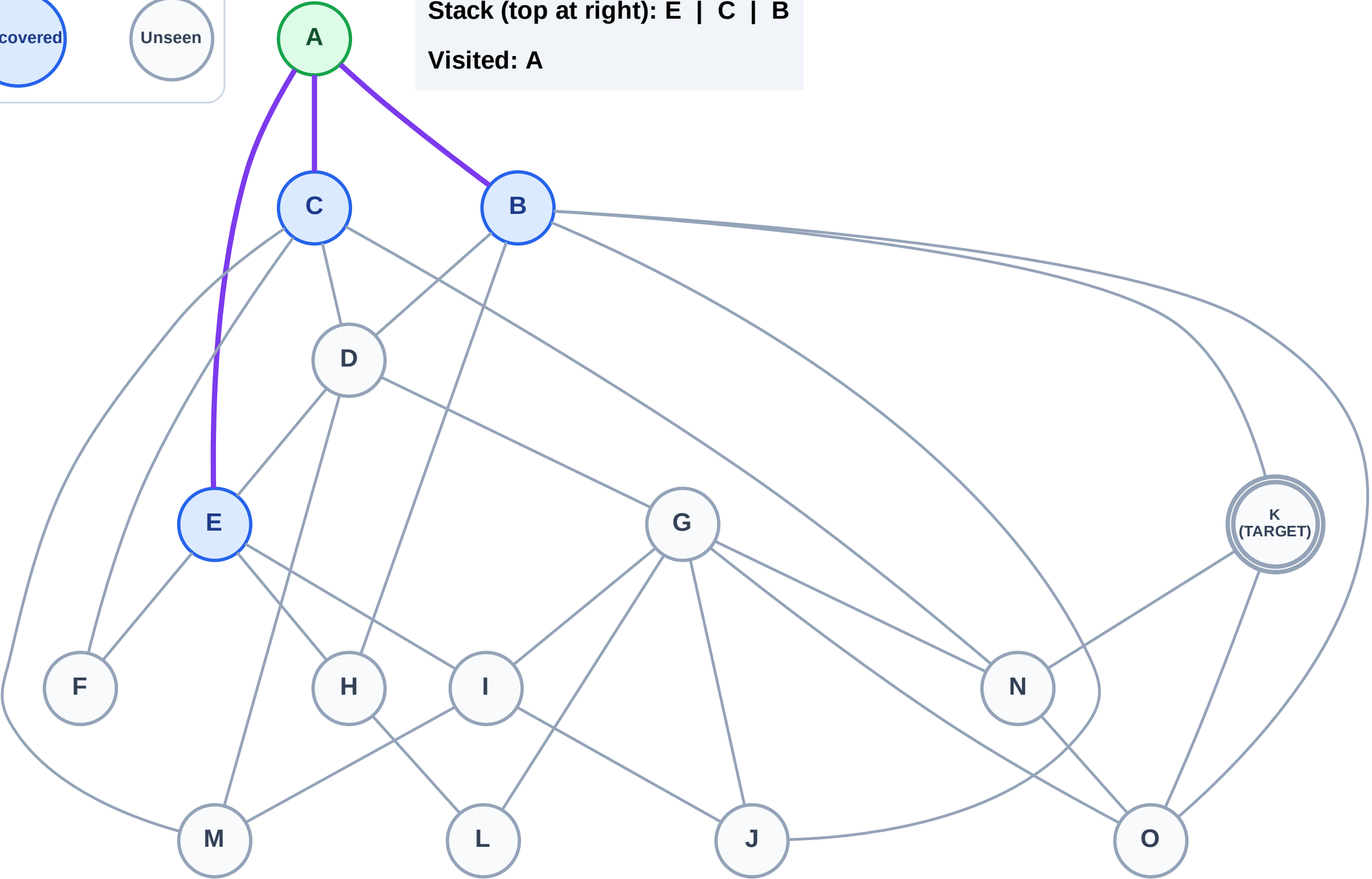
Processing

Discovered

Unseen

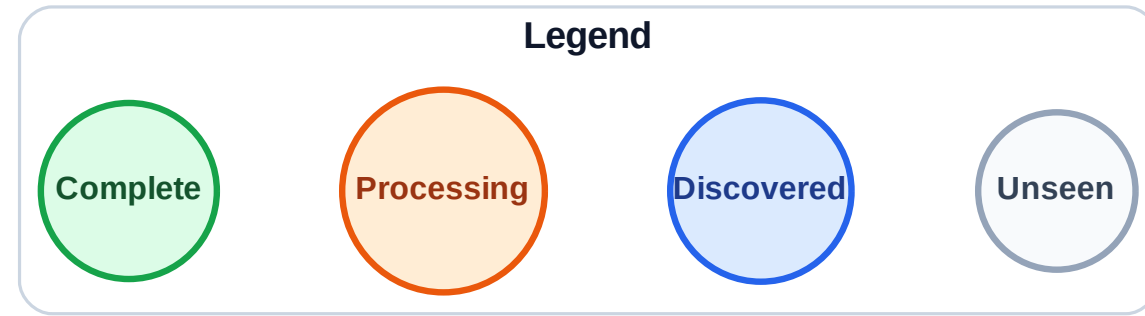
Stack (top at right): E | C | B

Visited: A

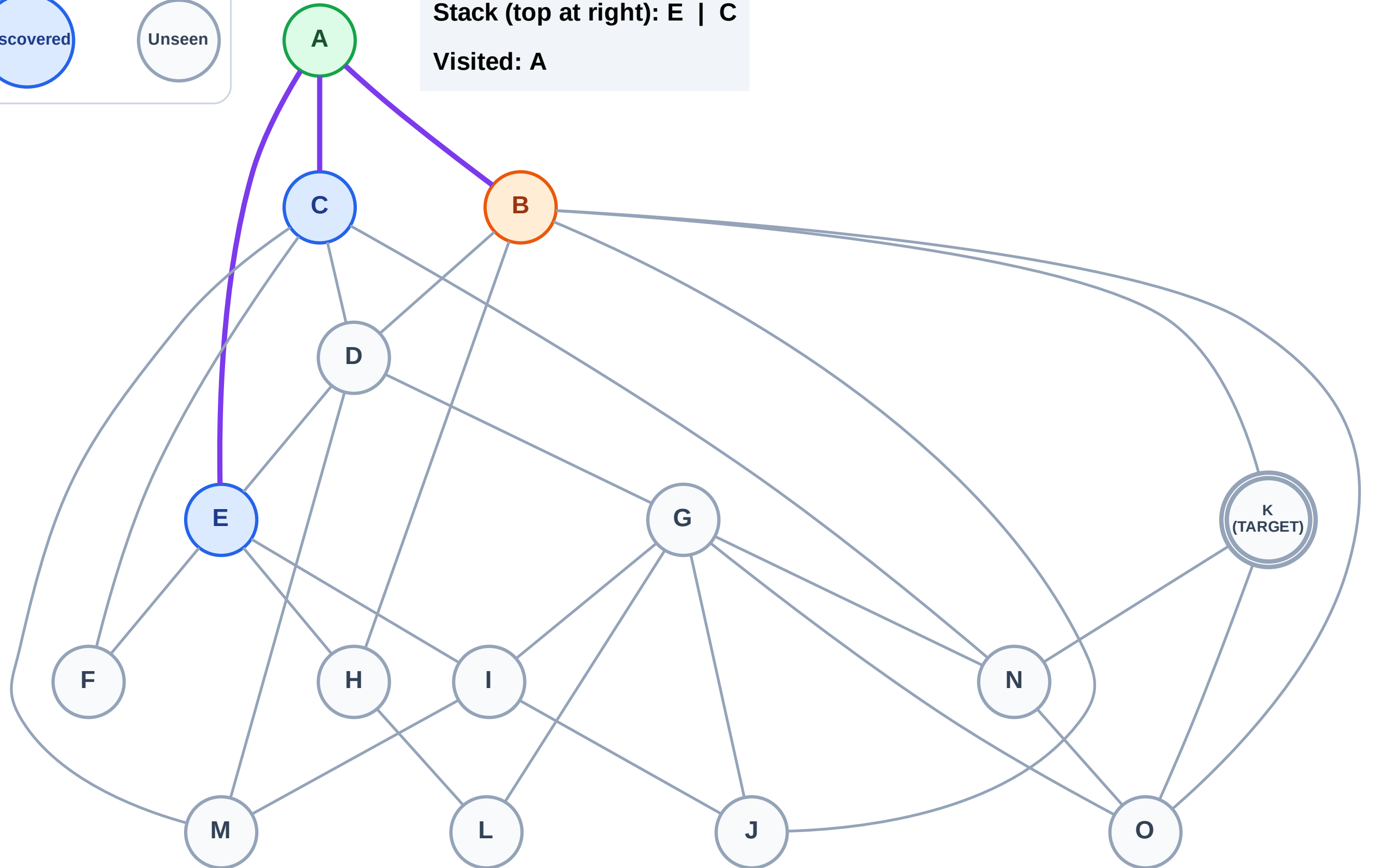


DFS Traversal

Step 4: Process node B



Stack (top at right): E | C
Visited: A



DFS Traversal

Step 5: Discover O, K, J, H, D from B

Legend

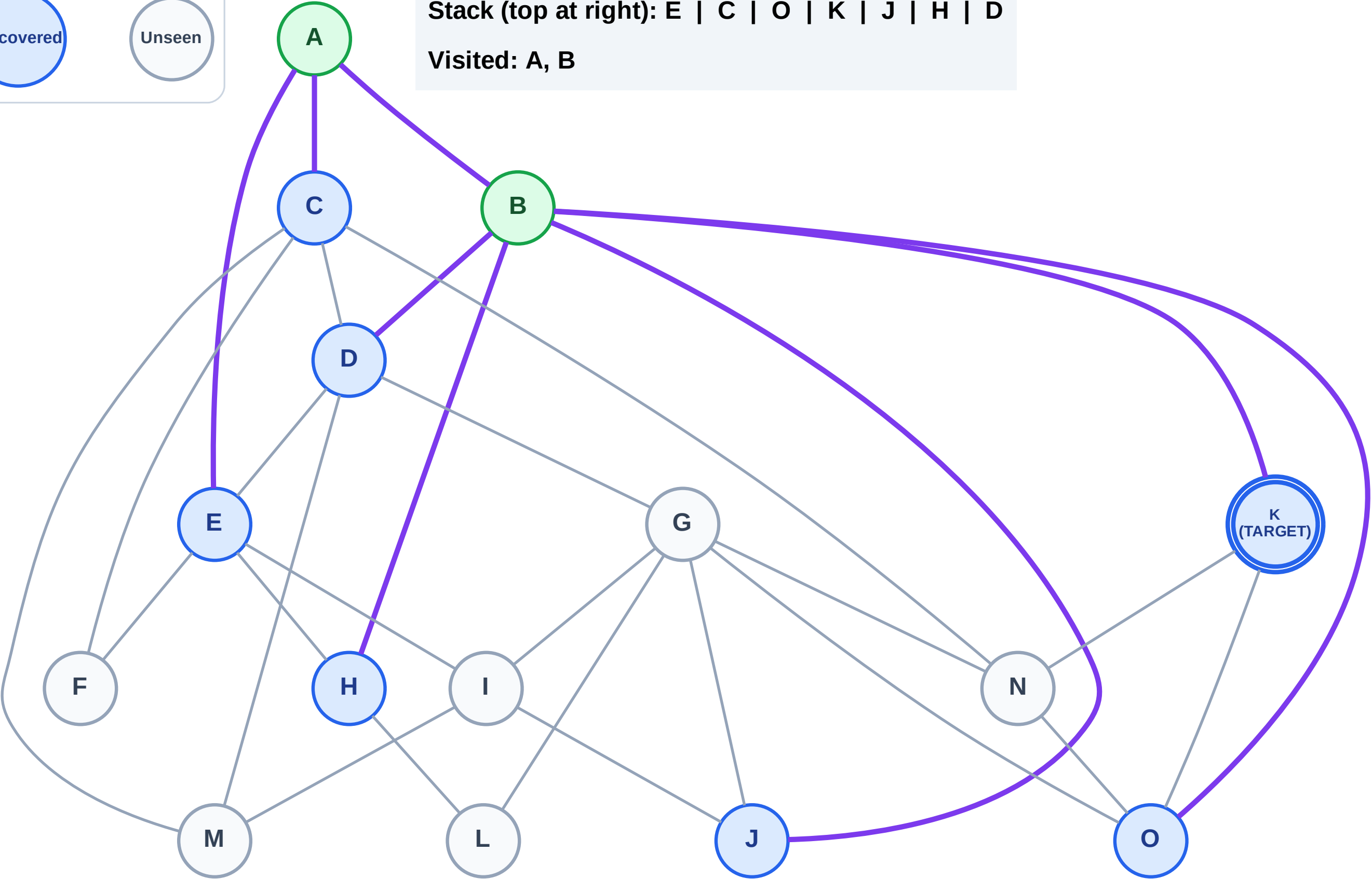
Complete

Processing

Discovered

Unseen

Stack (top at right): E | C | O | K | J | H | D
Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

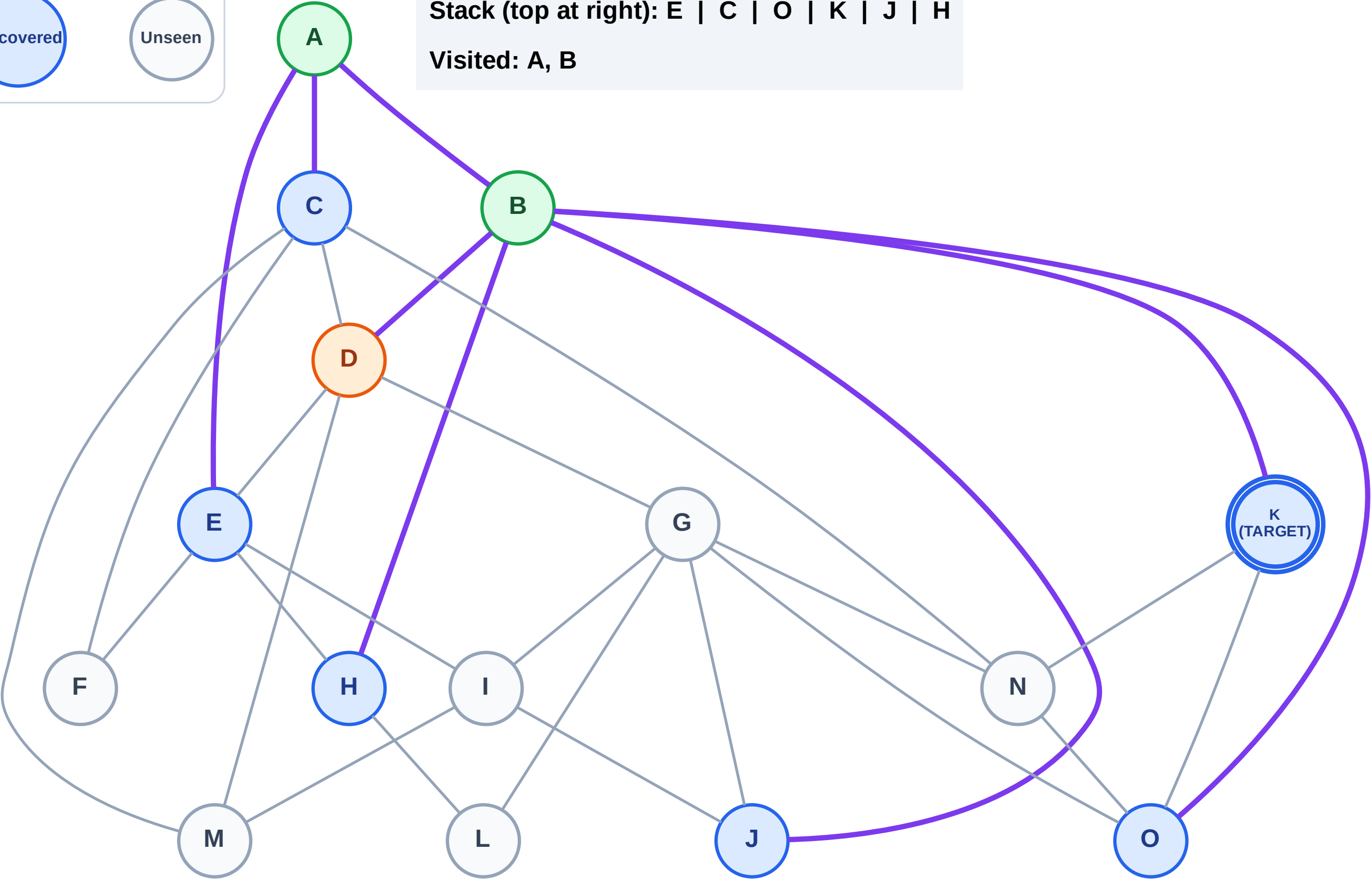
Processing

Discovered

Unseen

Stack (top at right): E | C | O | K | J | H

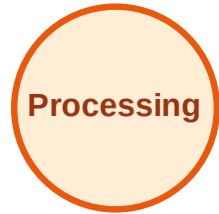
Visited: A, B



DFS Traversal

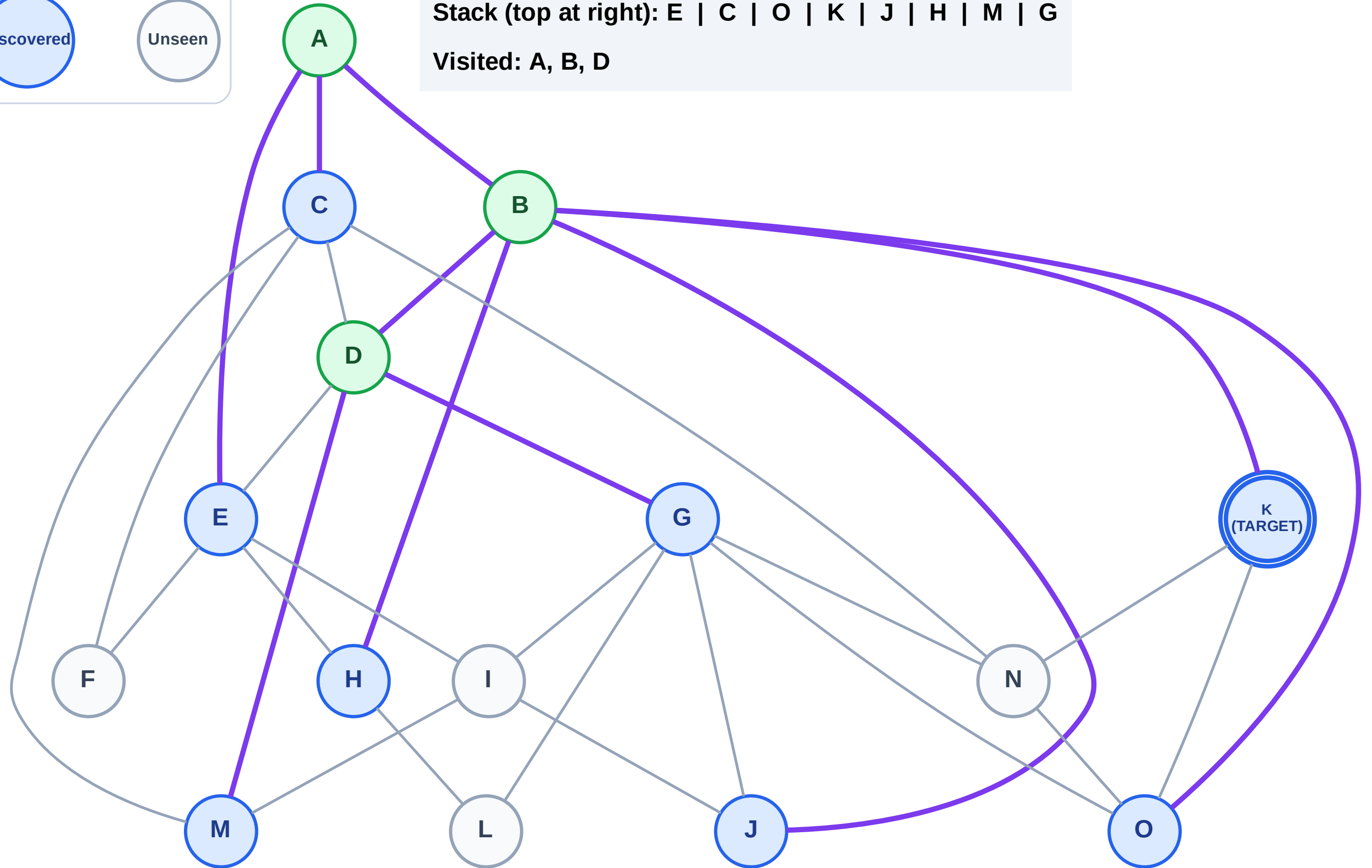
Step 7: Discover M, G from D

Legend



Stack (top at right): E | C | O | K | J | H | M | G

Visited: A, B, D



DFS Traversal

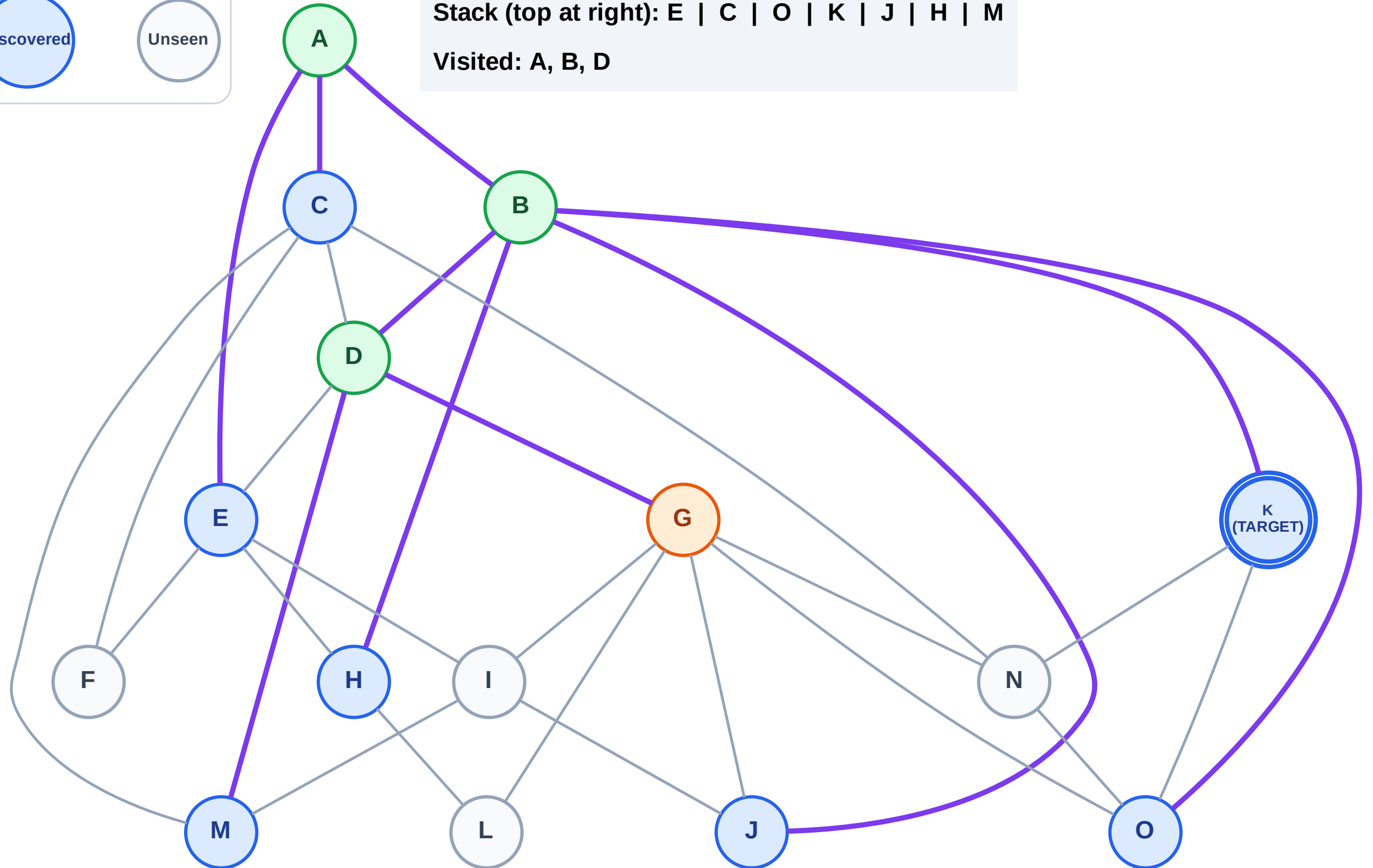
Step 8: Process node G

Legend



Stack (top at right): E | C | O | K | J | H | M

Visited: A, B, D



DFS Traversal

Step 9: Discover N, L, I from G

Legend

Complete

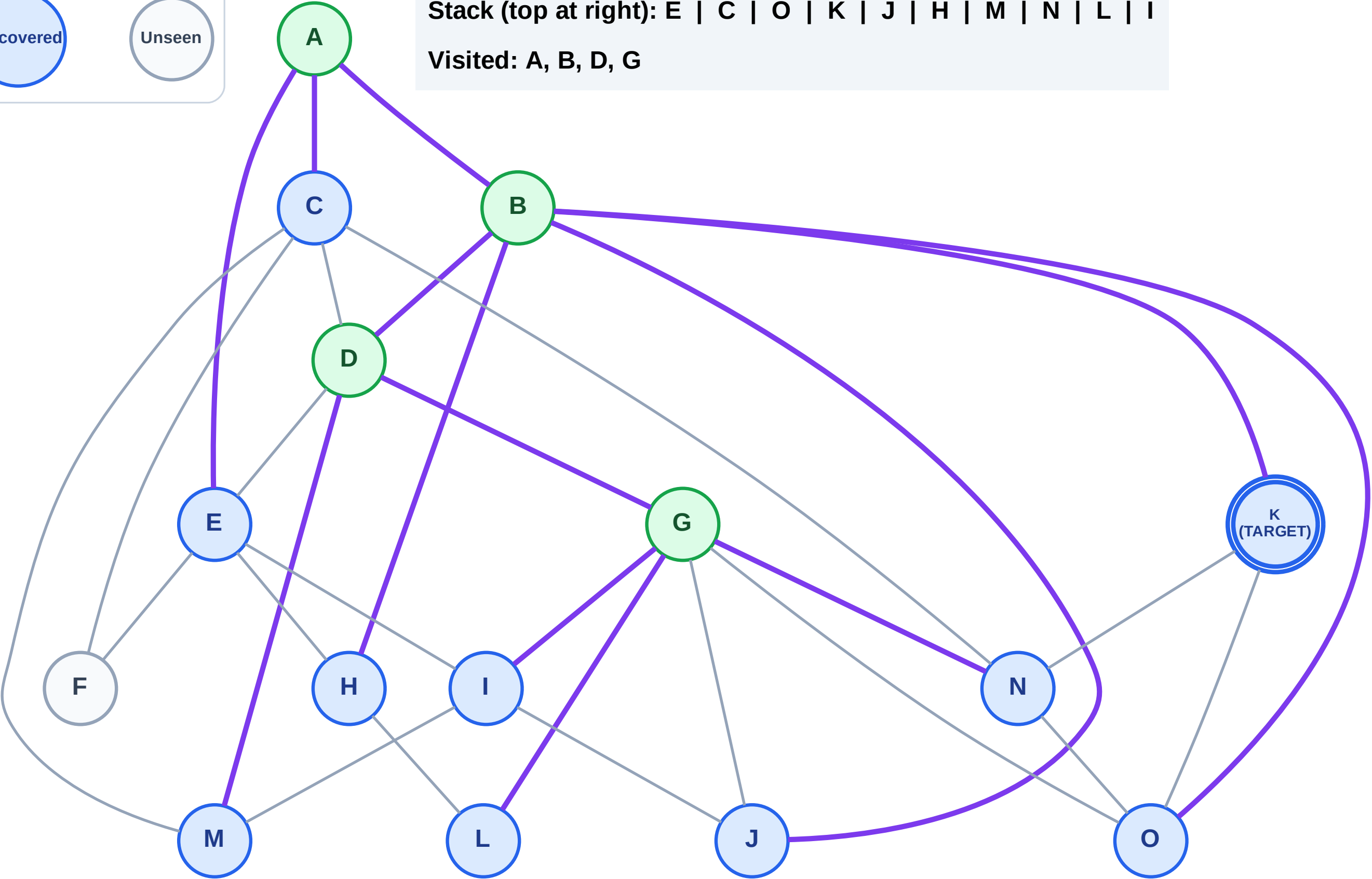
Processing

Discovered

Unseen

Stack (top at right): E | C | O | K | J | H | M | N | L | I

Visited: A, B, D, G



DFS Traversal

Step 10: Process node I

Legend

Complete

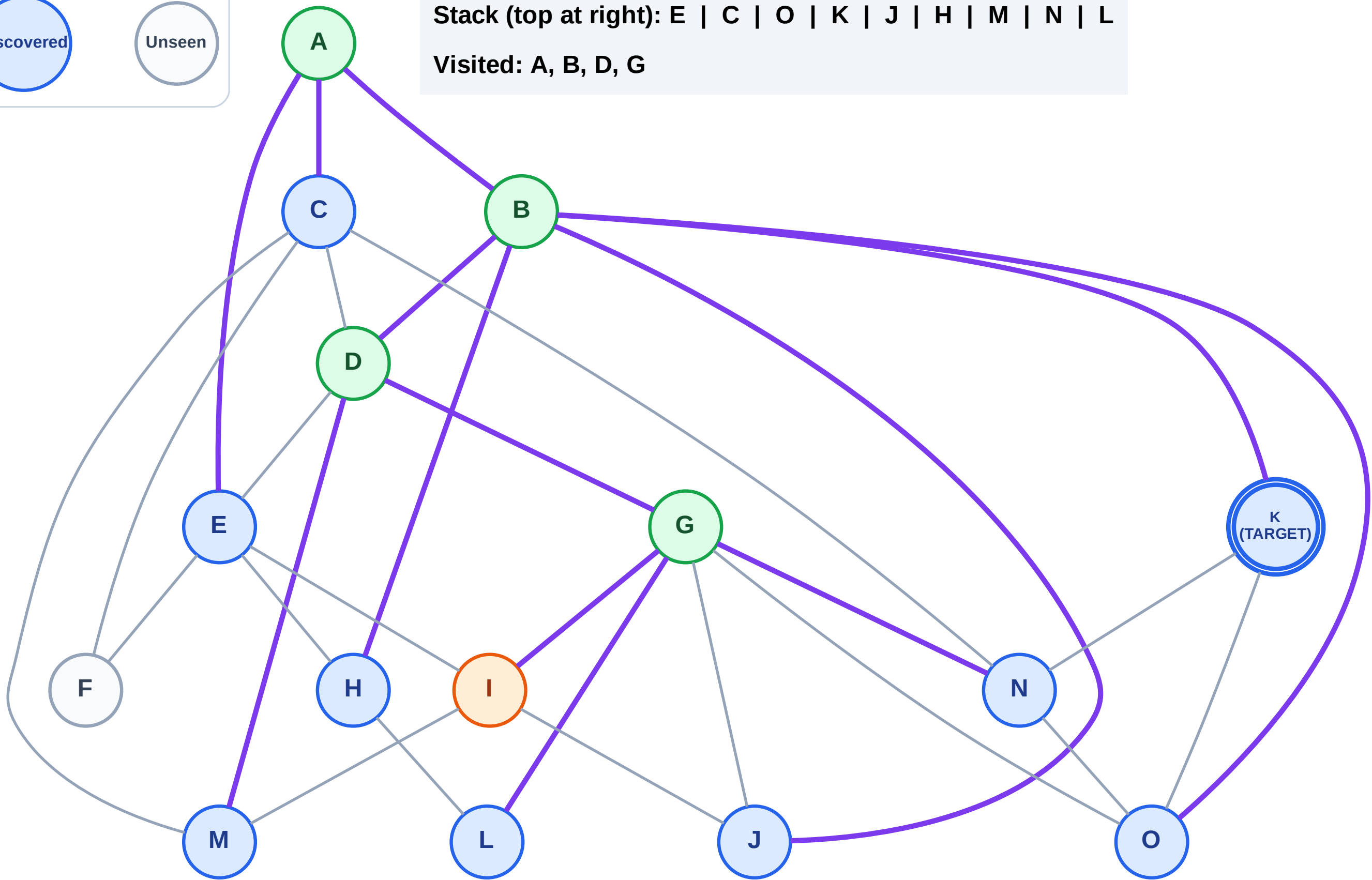
Processing

Discovered

Unseen

Stack (top at right): E | C | O | K | J | H | M | N | L

Visited: A, B, D, G



DFS Traversal

Step 11: No new neighbors from I

Legend

Complete

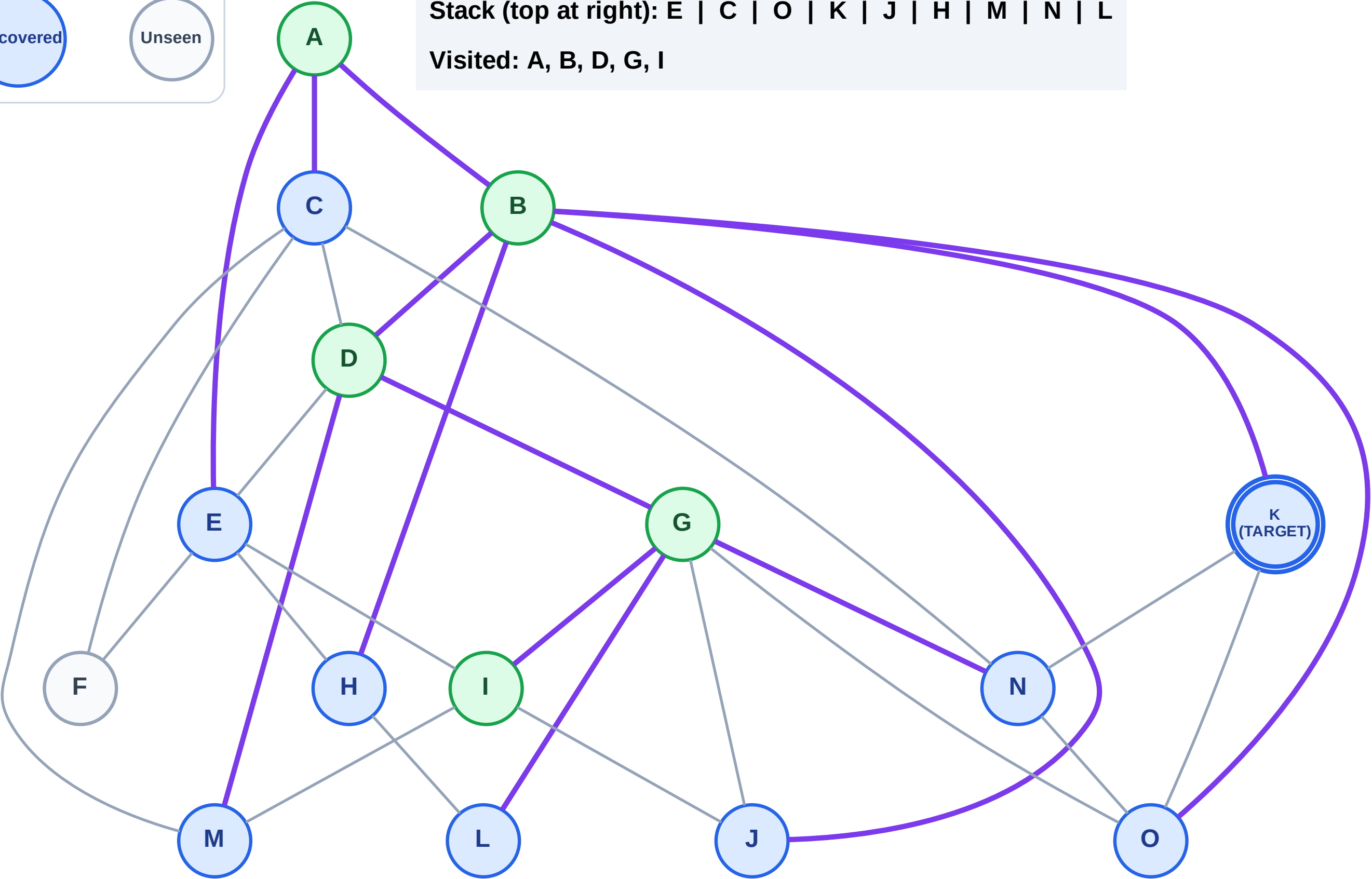
Processing

Discovered

Unseen

Stack (top at right): E | C | O | K | J | H | M | N | L

Visited: A, B, D, G, I



DFS Traversal

Step 12: Process node L

Legend

Complete

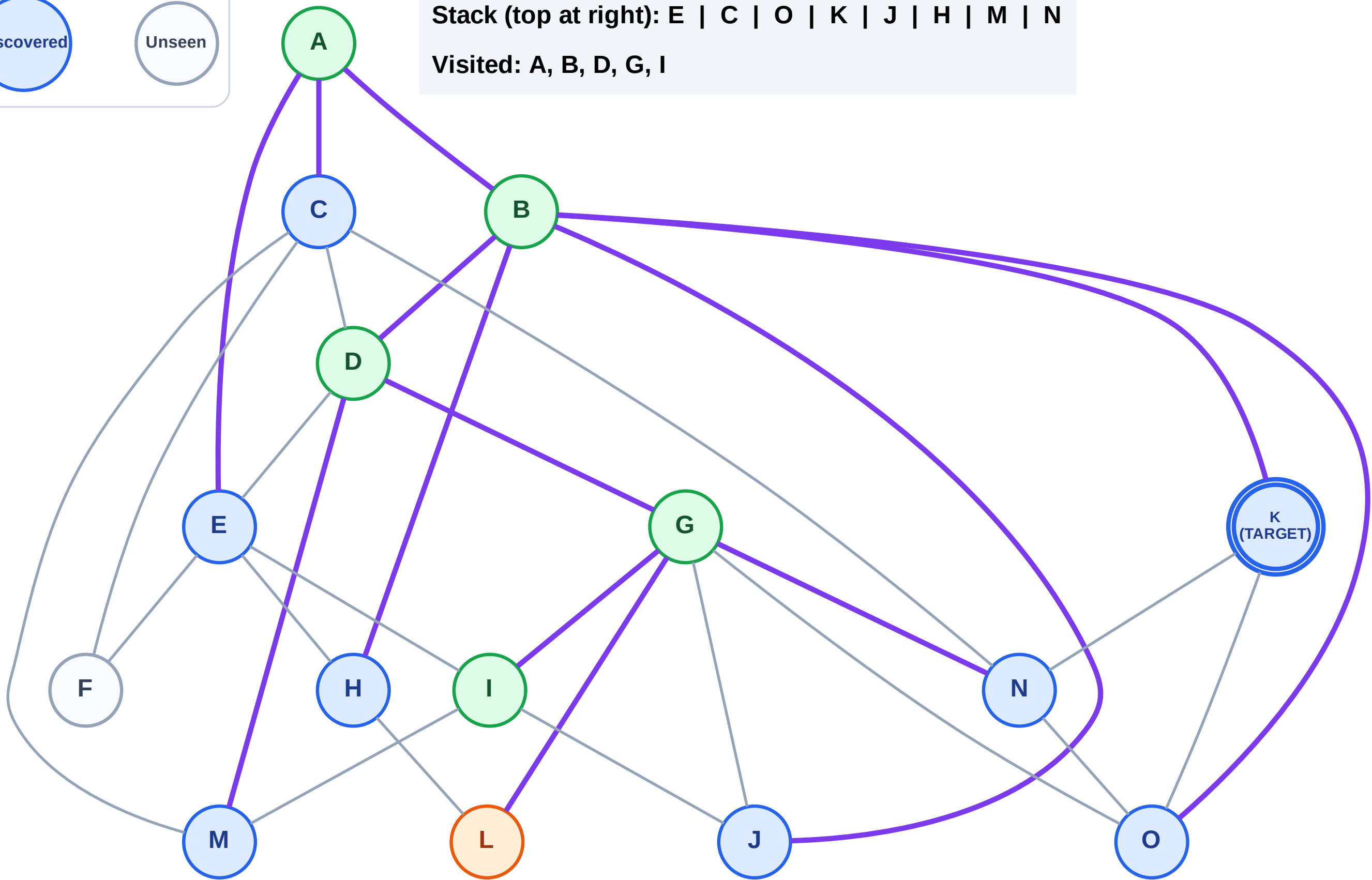
Processing

Discovered

Unseen

Stack (top at right): E | C | O | K | J | H | M | N

Visited: A, B, D, G, I



DFS Traversal

Step 13: No new neighbors from L

Legend

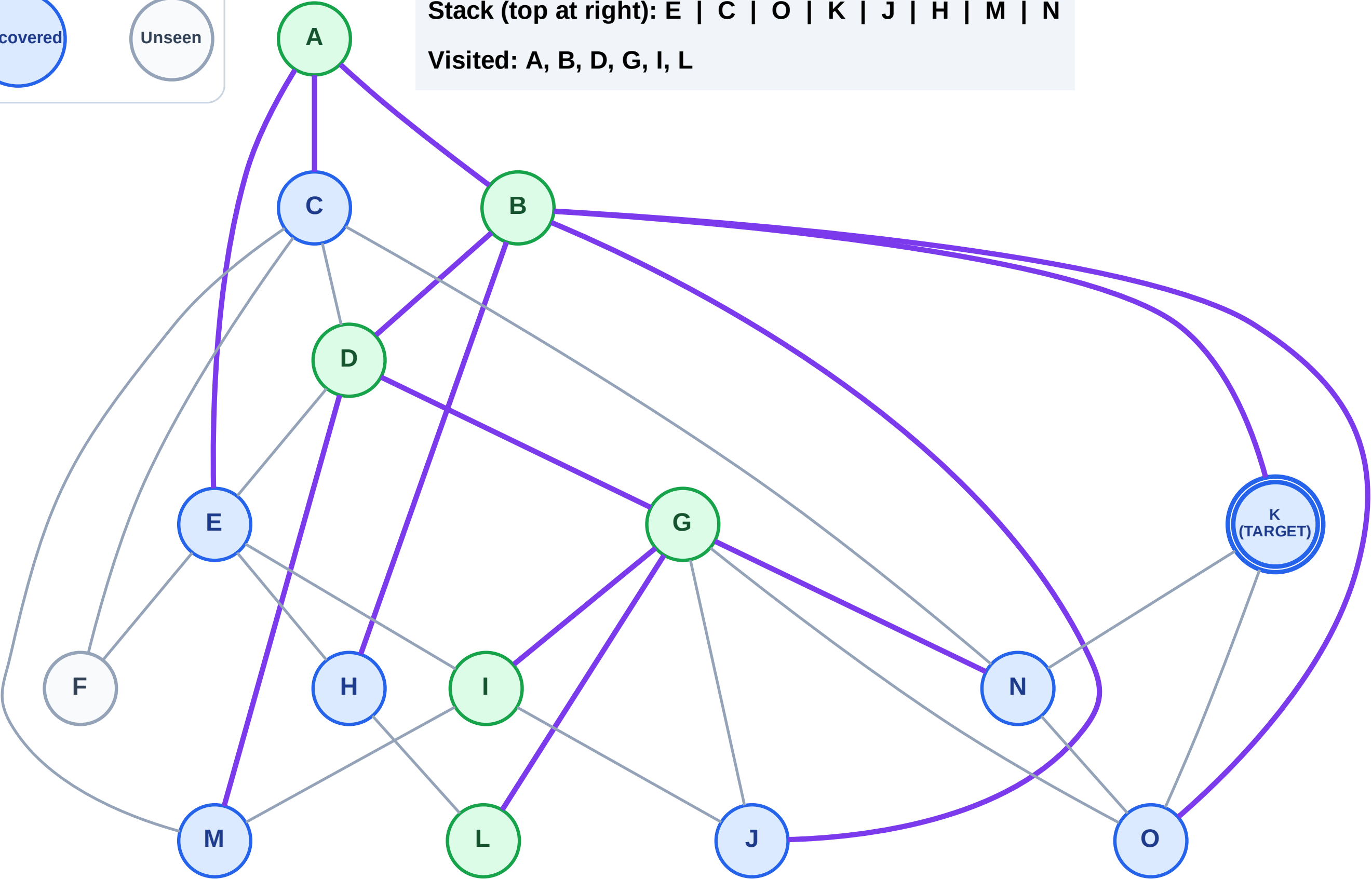
Complete

Processing

Discovered

Unseen

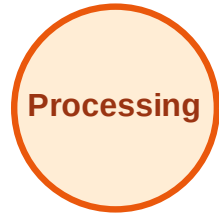
Stack (top at right): E | C | O | K | J | H | M | N
Visited: A, B, D, G, I, L



DFS Traversal

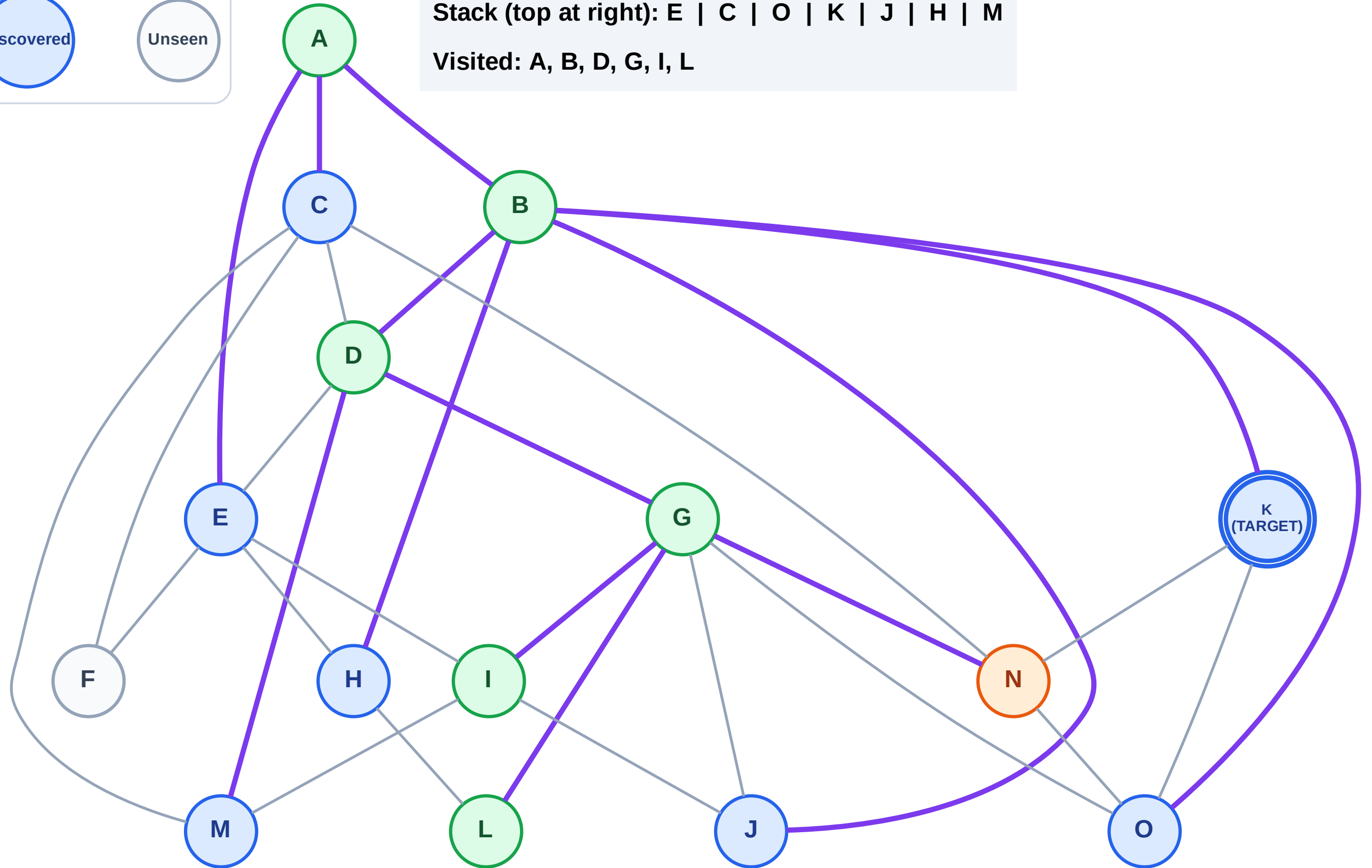
Step 14: Process node N

Legend



Stack (top at right): E | C | O | K | J | H | M

Visited: A, B, D, G, I, L



DFS Traversal

Step 15: No new neighbors from N

Legend

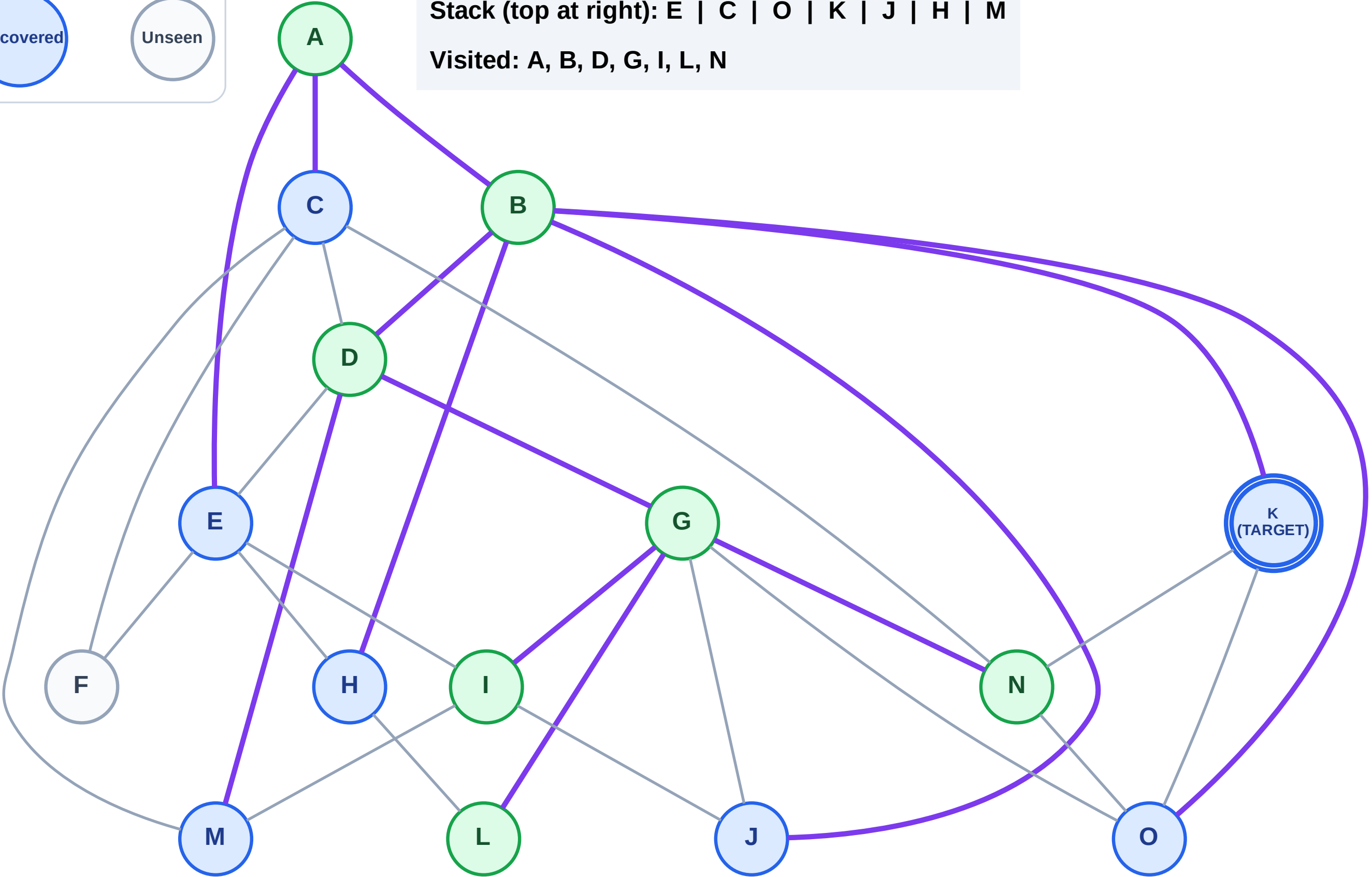
Complete

Processing

Discovered

Unseen

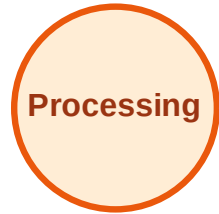
Stack (top at right): E | C | O | K | J | H | M
Visited: A, B, D, G, I, L, N



DFS Traversal

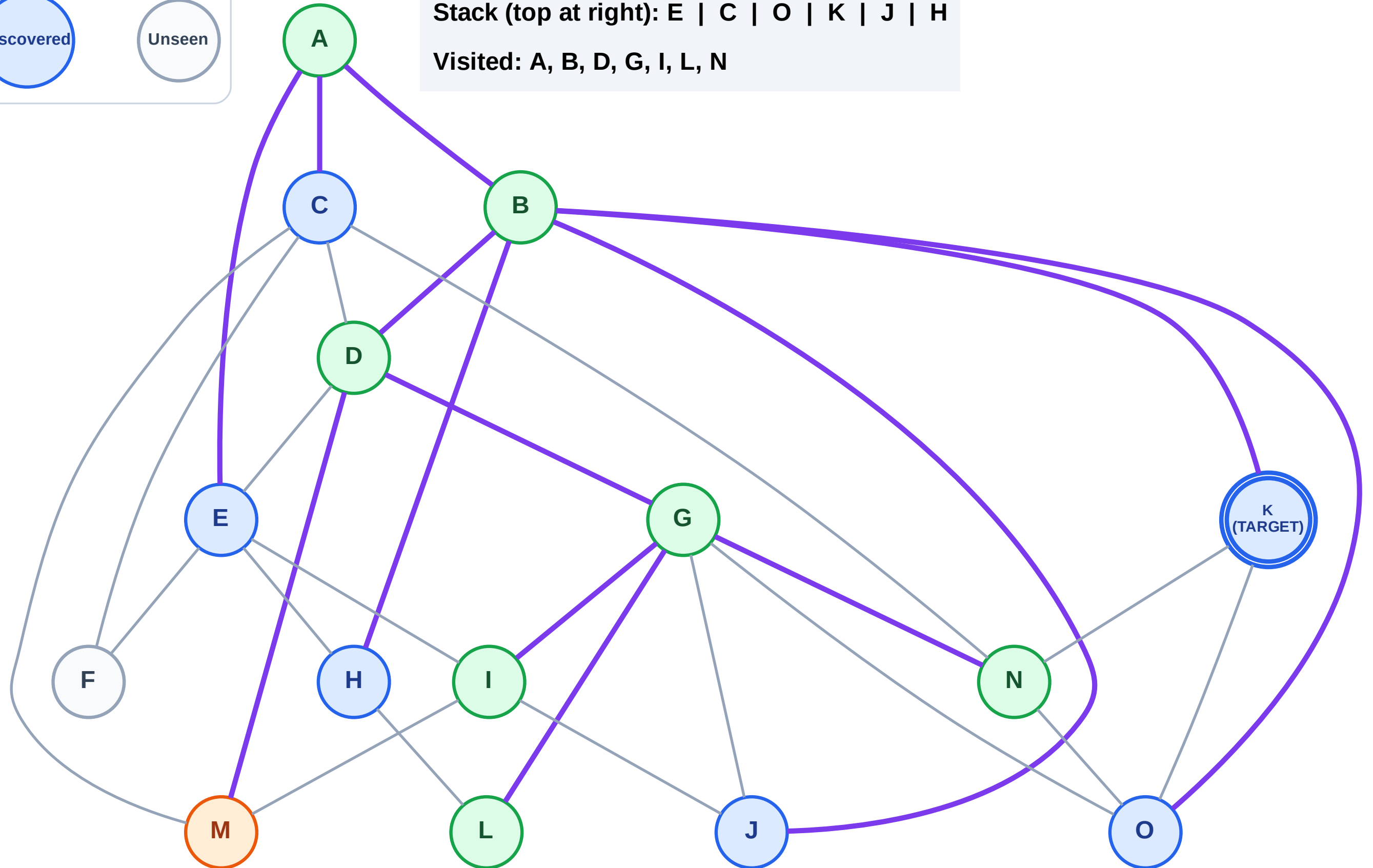
Step 16: Process node M

Legend



Stack (top at right): E | C | O | K | J | H

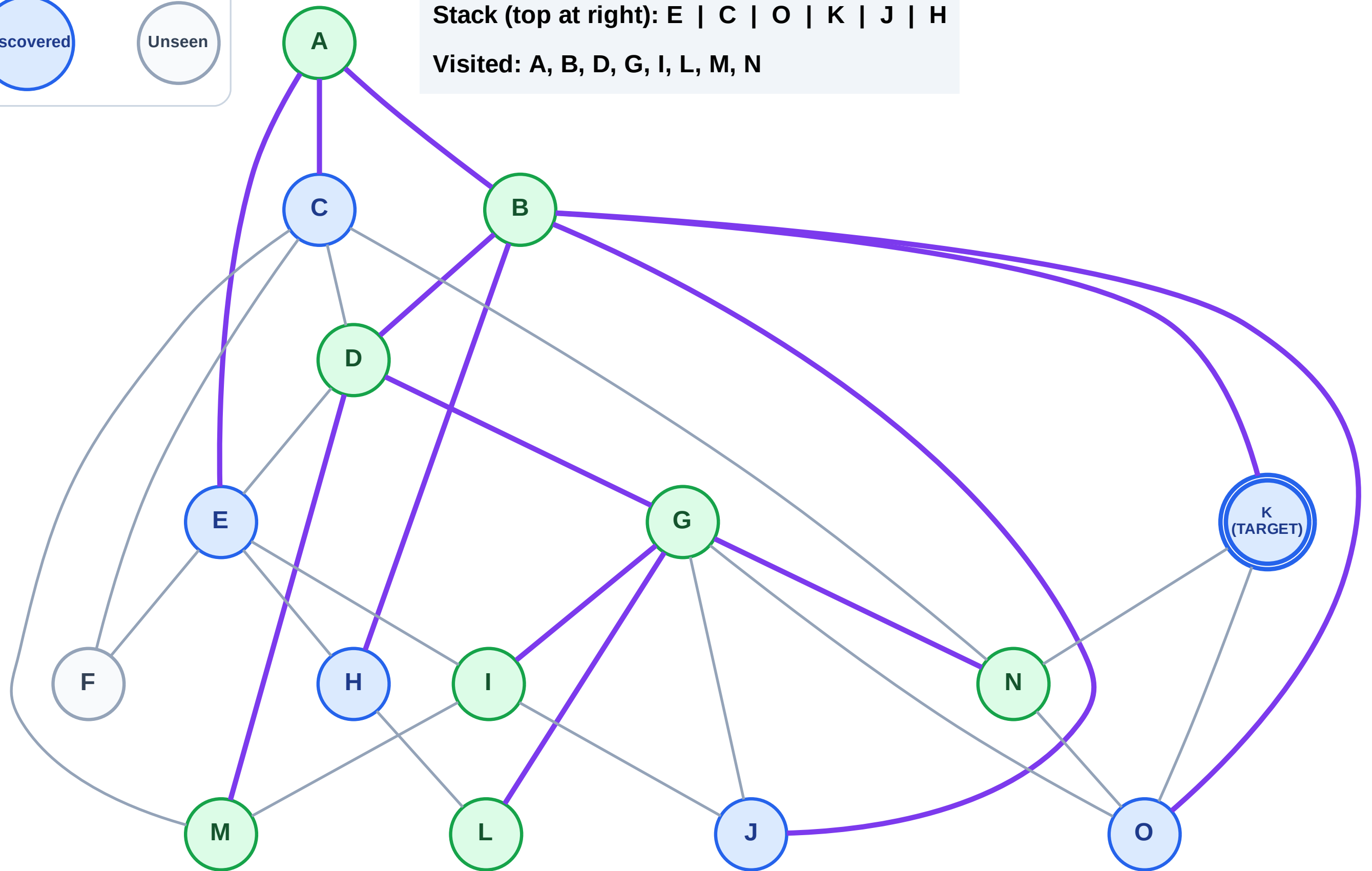
Visited: A, B, D, G, I, L, N



Step 17: No new neighbors from M

Stack (top at right): E | C | O | K | J | H

Visited: A, B, D, G, I, L, M, N



DFS Traversal

Step 18: Process node H

Legend

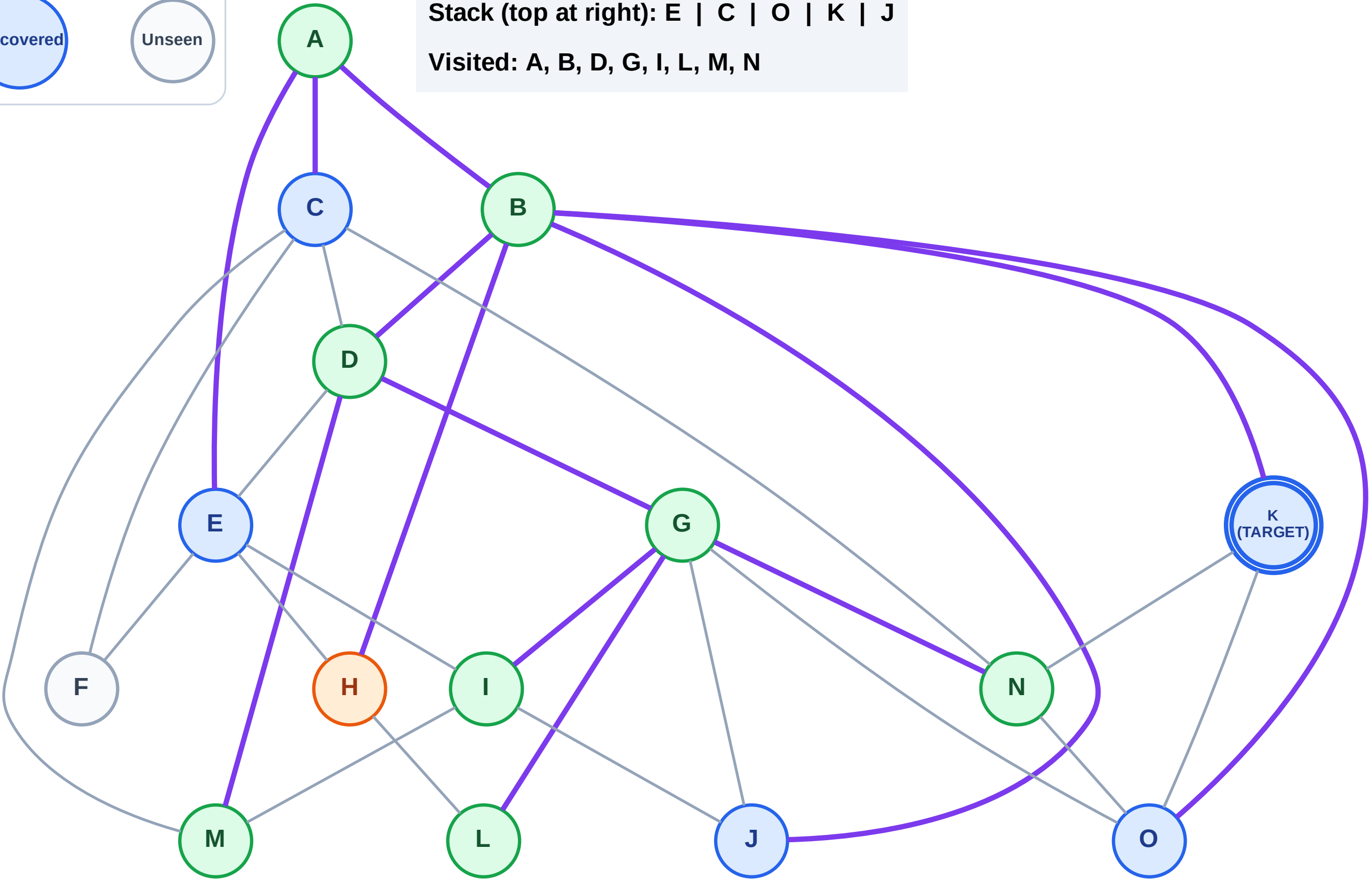
Complete

Processing

Discovered

Unseen

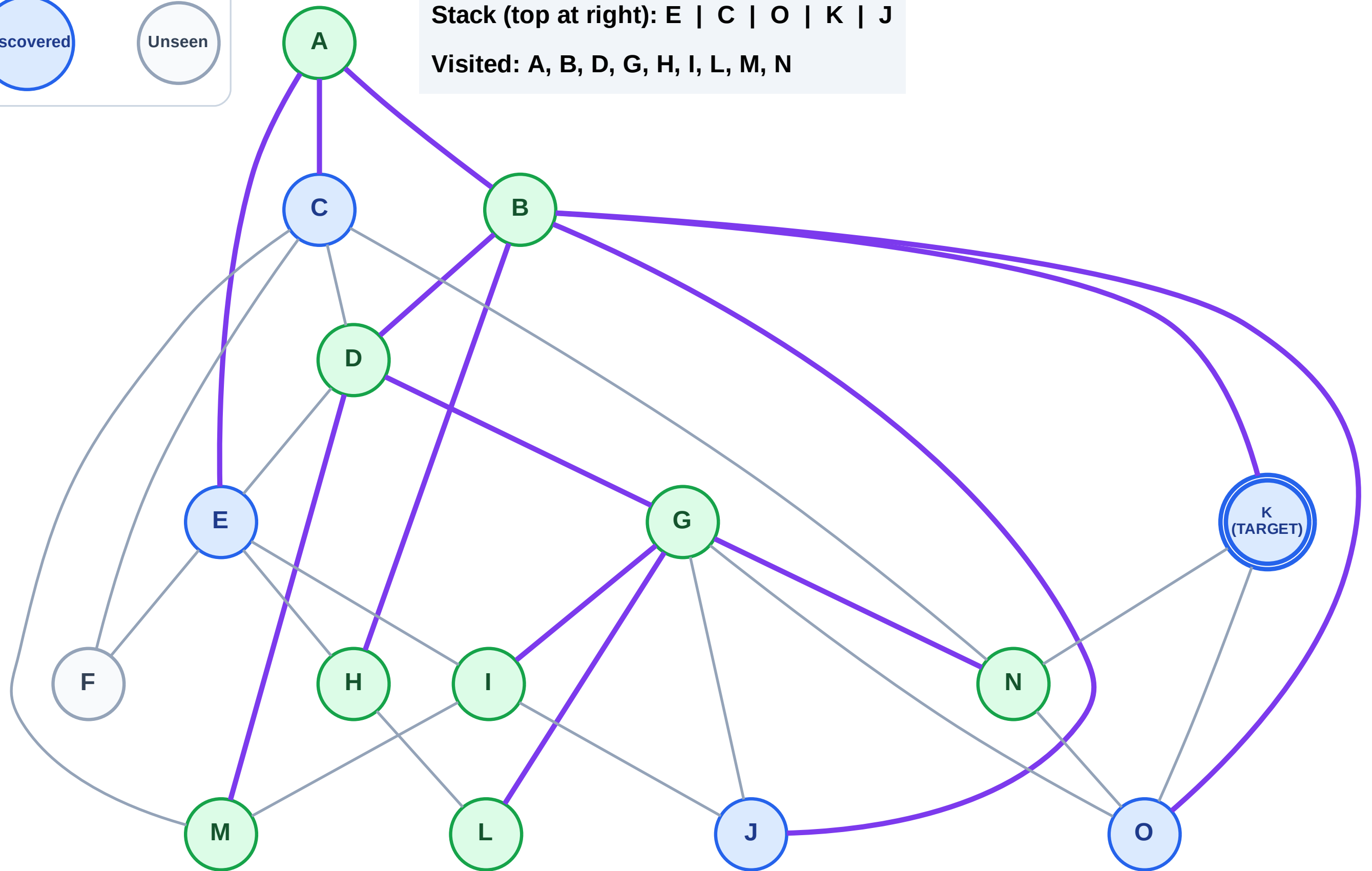
Stack (top at right): E | C | O | K | J
Visited: A, B, D, G, I, L, M, N



Step 19: No new neighbors from H

Stack (top at right): E | C | O | K | J

Visited: A, B, D, G, H, I, L, M, N



DFS Traversal

Step 20: Process node J

Legend

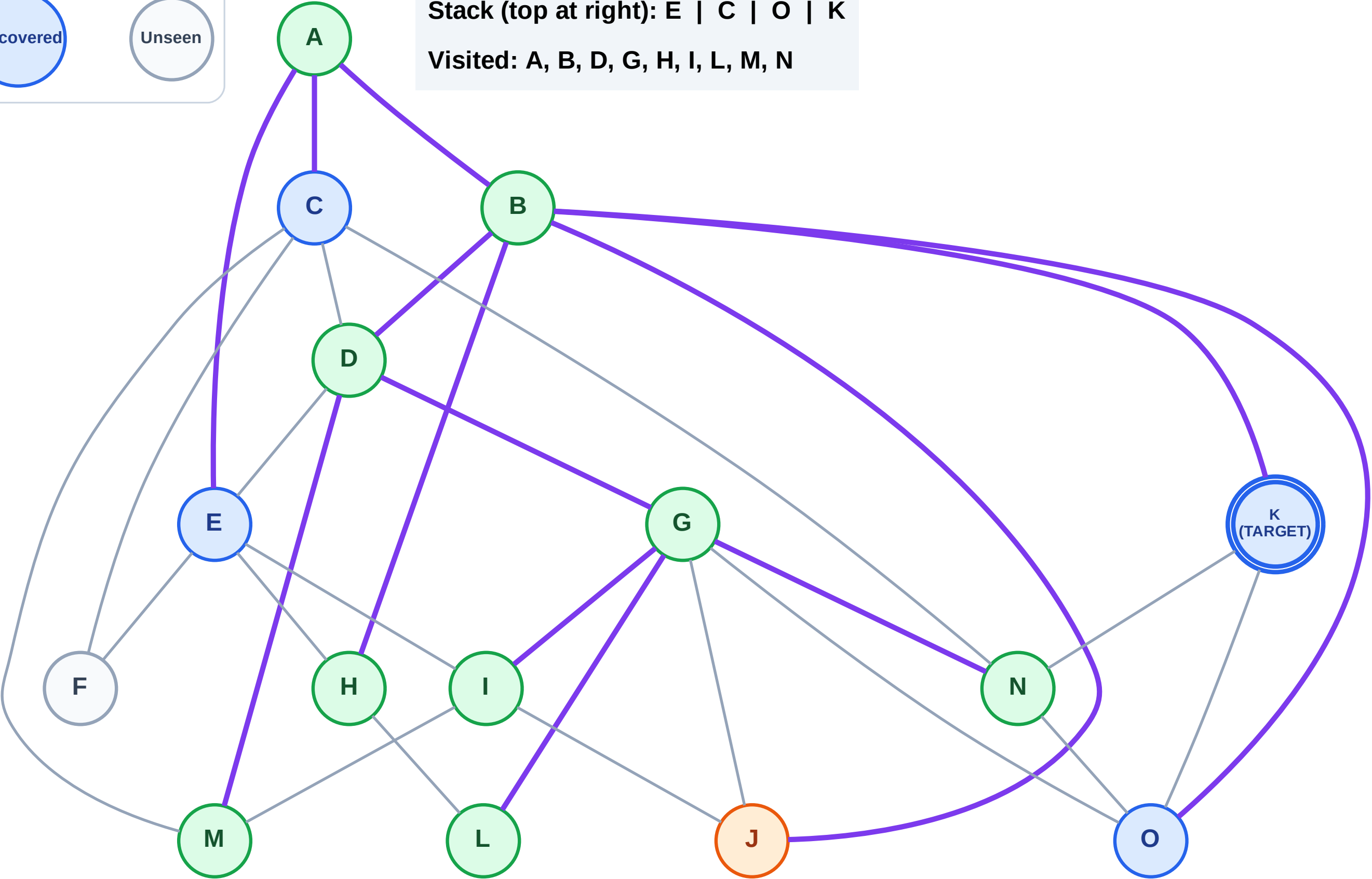
Complete

Processing

Discovered

Unseen

Stack (top at right): E | C | O | K
Visited: A, B, D, G, H, I, L, M, N



DFS Traversal

Step 21: No new neighbors from J

Legend

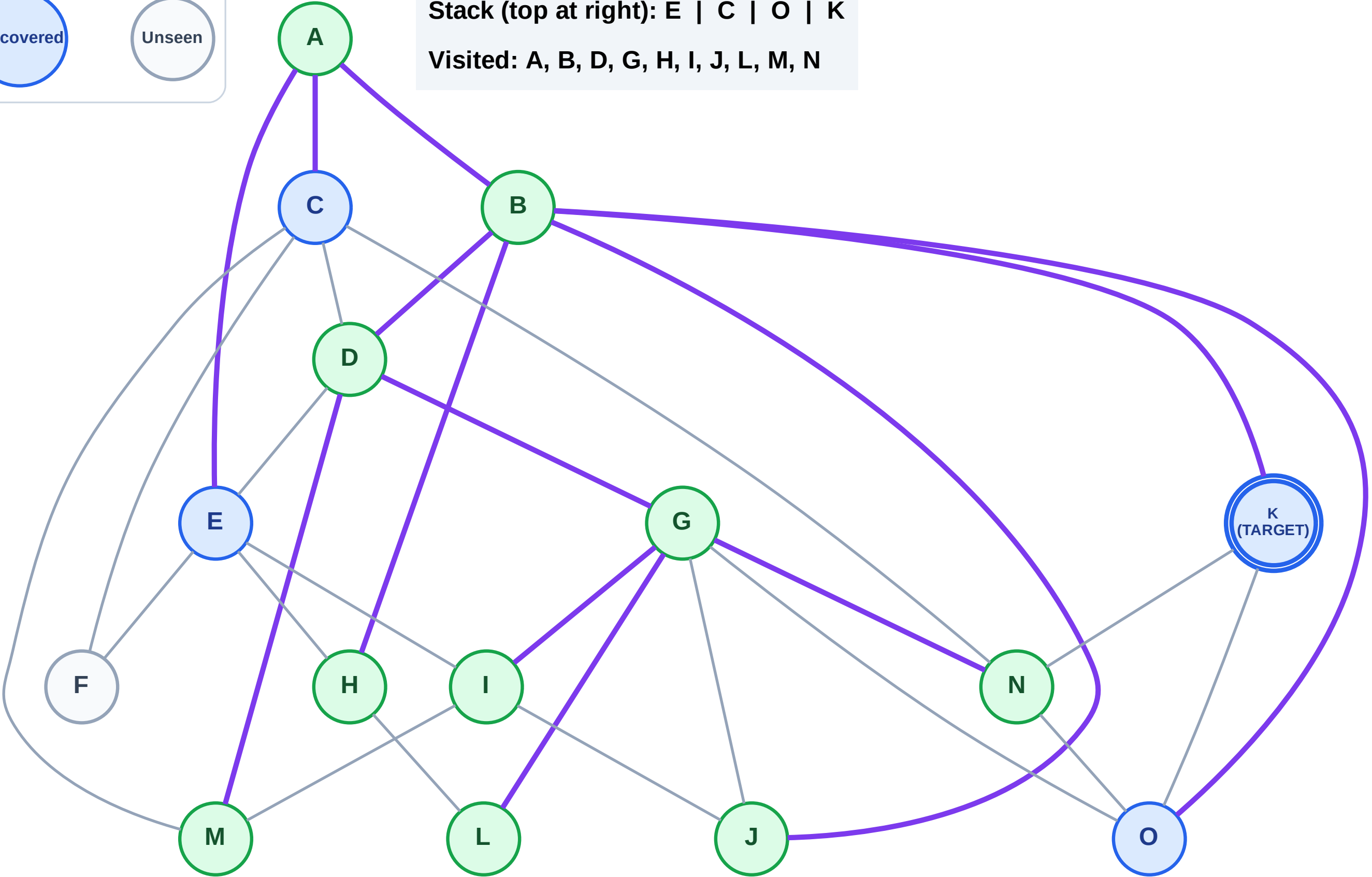
Complete

Processing

Discovered

Unseen

Stack (top at right): E | C | O | K
Visited: A, B, D, G, H, I, J, L, M, N



DFS Traversal

Step 22: Process node K

Legend

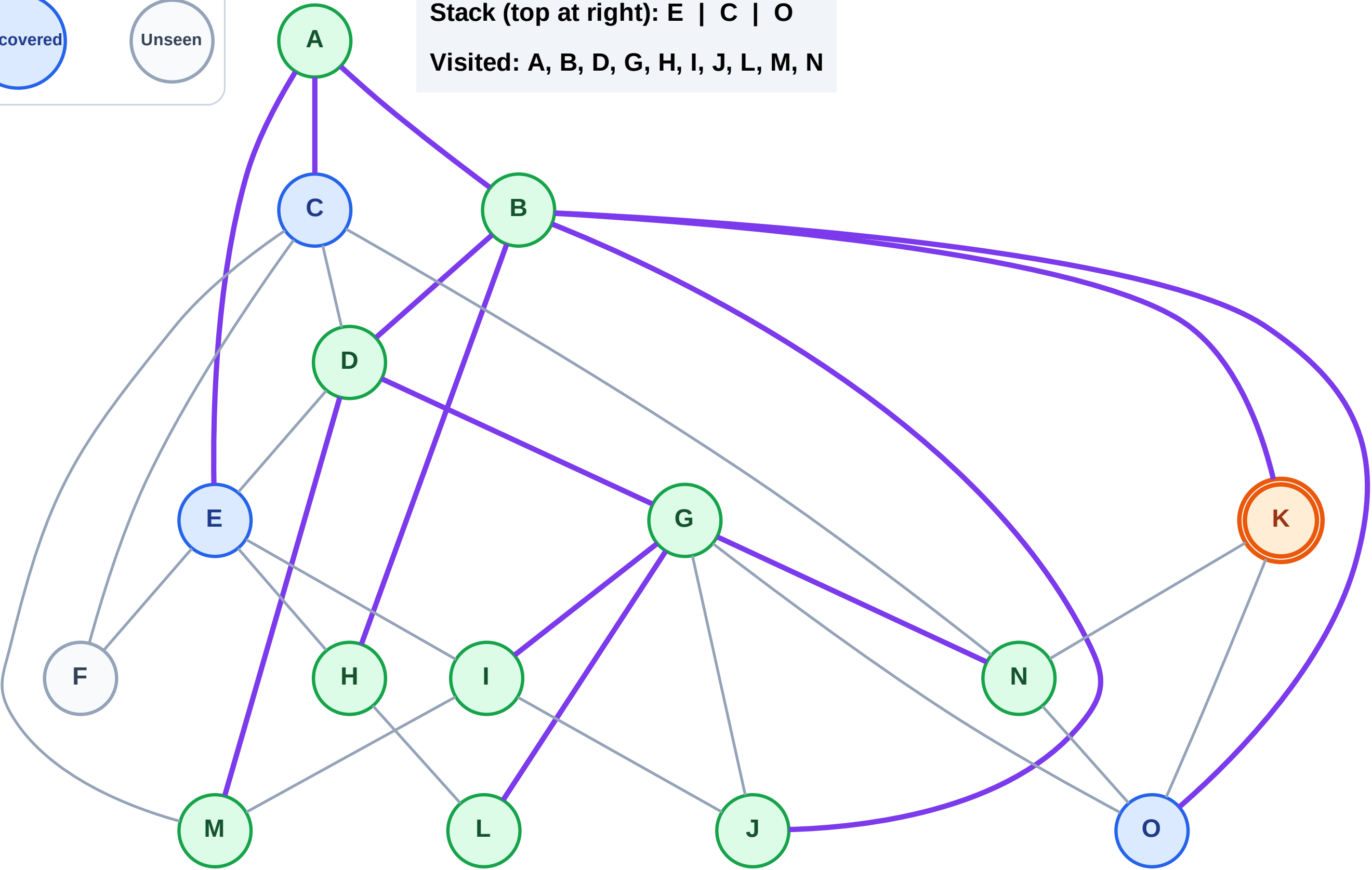
Complete

Processing

Discovered

Unseen

Stack (top at right): E | C | O
Visited: A, B, D, G, H, I, J, L, M, N



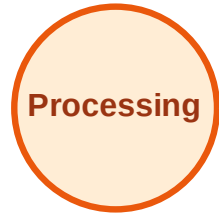
DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): E | C | O

Visited: A, B, D, G, H, I, J, K, L, M, N

